

Computer Networks & Cybersecurity — ISRO CSE Revision Notes

Concise notes + relevant exam questions, compressed from the KnowledgeGate CN & Cybersecurity module.

1. Introduction to Computer Networks

1.1 Basics, Goals & Applications

- **Network** = autonomous computers interconnected by one technology to share resources. Internet = "network of networks."
- **Goals:** Resource sharing, Communication, Reliability (alternate paths), Cost reduction (centralize hardware), Scalability, Remote access.
- **Applications:** Resource sharing, Interpersonal communication (email/VoIP/video conf), Remote info access (WWW), Distributed/Cloud computing, E-commerce, Interactive entertainment (VoD, gaming).
- **Data communication = 5 components:** Message, Sender, Receiver, Transmission medium, Protocol.
- **4 characteristics of effective data communication:** Delivery, Accuracy, Timeliness, **Jitter** (variation in packet arrival time — **TRAP:** jitter $\neq$ delay, it's the *variation* in delay).
- **Network criteria:** Performance, Reliability, Security.

1.2 Evolution of Networking

Standalone computers → ARPANET (1969) → TCP/IP adoption (1983) → NSFNET (1986) → Commercial Internet + WWW (1990s) → Modern Internet

- **ARPANET** (1969, DARPA/US DoD): first operational **packet-switching** network; goal = survive partial failure (nuclear attack resilience); first link UCLA→SRI.
- **1983:** ARPANET migrates from NCP to **TCP/IP** — this is considered the true birth of "the Internet."
- **NSFNET** (1986): high-speed academic backbone connecting 5 US supercomputing centers; replaced ARPANET; decommissioned **1995** once commercial ISPs took over.
- **WWW** (1991, Tim Berners-Lee): added HTTP/HTML, made Internet browsable.
- **TRAP:** ARPANET $\neq$ Internet. ARPANET is the *ancestor*; "Internet" formally begins with TCP/IP standardization (1983).

1.3 Transmission Modes & Connection Types

Mode	Direction	Example
Simplex	One-way only	Radio, keyboard→CPU
Half-Duplex	Both ways, not simultaneous	Walkie-talkie
Full-Duplex	Both ways, simultaneous (= 2 half-duplex)	Telephone

- **Point-to-point:** dedicated link between exactly 2 devices.
- **Multipoint (multidrop):** single link shared by more than 2 devices.

1.4 Network Topologies

Topology	Links Needed	Key Advantage	Key Disadvantage
Mesh	$n(n-1)/2$ dedicated links	No traffic issues, robust, private, easy fault ID	Expensive, hard to install/wire
Star	n links to central hub	Easy install/reconfig/fault-ID, robust to single-link failure	Hub is single point of failure
Bus	1 backbone cable	Least cabling, easy install	Break in cable halts all; hard fault isolation
Ring	n links (closed loop)	Simple install, easy fault isolation	One device failure can break the ring
Tree	Star-of-Stars (Bus+Star hybrid)	Scalable, fault isolated per branch	Backbone failure kills whole network
Hybrid	Combination of ≥ 2 topologies	Flexible, fault-tolerant across sub-nets	Complex & expensive to design

TRAP: Mesh topology needs $n(n-1)/2$ links (undirected/duplex) — a common numerical question. For $n=5 \rightarrow 10$ links; each node needs $(n-1)$ ports.

1.5 Types of Networks (by scale)

Feature	PAN	LAN	MAN	WAN
Range	~10 m	few km (building/campus)	5–50 km (city)	Unlimited (global)
Standard	Bluetooth (802.15.1)	Ethernet (802.3) / Wi-Fi (802.11)	Fiber/Microwave	Satellite, undersea cable, leased lines
Ownership	Personal	Private (single org)	Public/Private (ISP/Govt)	Mixed, no single owner
Speed	Moderate	Very high (100 Mbps–10 Gbps)	Moderate–high	Low (vs LAN) — higher latency
Example	Bluetooth headset	Office/School LAN	Cable TV network, city Wi-Fi	Internet, MNC WAN, 4G/5G

1.6 Switching Techniques

Aspect	Circuit Switching	Packet Switching
Path	Dedicated, reserved before transfer	No fixed path; dynamic per packet
Connection	Connection-oriented	Connectionless (per packet)
Bandwidth	Reserved (wasted when idle)	Shared dynamically — better utilization
Delay	Setup delay, then constant	Variable (queuing/congestion); no setup delay
Order	In-order (same path)	May arrive out-of-order — needs reassembly
Example	Telephone network	Internet

- **Message Switching:** whole message (not packets) stored-and-forwarded hop-by-hop; high delay/memory; not real-time (legacy: telegram, early email).
- Circuit switching phases: **Connection establishment** → **Data transfer** → **Connection release**.
- **TRAP:** Packet switching ⇒ possible packet loss/out-of-order delivery; circuit switching ⇒ guaranteed bandwidth but wastes it when idle.

1.7 X-Series Standards (ITU-T, for Public Data Networks)

- **X.21** — Physical layer interface for **circuit-switched** networks (DTE↔DCE signalling).
- **X.25** — classic **packet switching**, connection-oriented (virtual circuit); maps to OSI L1–L3: **X.21 (Physical)** → **LAPB (Data Link)** → **PLP (Network)**. Hop-by-hop error checking ⇒ reliable but slow; superseded by Frame Relay / TCP/IP.

1.8 Transmission Media

Guided (wired): Twisted pair (telephone), Coaxial (cable TV), Fiber optic (light, total internal reflection, up to ~1600 Gbps, immune to EM noise, unidirectional light).

Unguided (wireless):

| Feature | Radio Waves | Microwaves | Infrared |

| ---|---|---|---|

| Frequency | 3 kHz–1 GHz | 1 GHz–300 GHz | 300 GHz–400 THz |

| Directionality | Omnidirectional | Unidirectional (Line-of-Sight) | Line-of-Sight |

| Penetrates walls? | Yes | No | No (→ secure, indoor use) |

| Use | AM/FM radio | Satellite, point-to-point towers, cellular backhaul | TV remote, wireless mouse |

TRAP: Microwave requires precise antenna alignment (LOS) and is blocked by obstacles — often confused with radio waves, which penetrate buildings.

1.9 OSI Model — 7 Layers

Layer	Core Job	Adds
7. Application	User-facing network services (mail, FTM, directory)	—
6. Presentation	Translation, Encryption , Compression	—
5. Session	Dialog control, synchronization checkpoints	—
4. Transport	Process-to-process, segmentation, flow/error control	Port (Service-point) address
3. Network	End-to-end packet delivery across networks, routing	Logical (IP) address
2. Data Link	Node-to-node reliable transfer, framing, MAC access control, flow/error control	Physical (MAC) address
1. Physical	Raw bit transmission, signaling, topology, transmission mode	—

- Mnemonic: **All People Seem To Need Data Processing** (L7→L1).
- **TRAP:** Encryption/compression is a **Presentation layer** job, not Application layer. Error control appears at BOTH Data Link (node-to-node) and Transport (end-to-end) — a classic "which layer does X" trap.
- OSI is a **reference model**, not a protocol itself (contrast with TCP/IP, which is an actual protocol suite).

Questions

Q1. How many dedicated links are required in a mesh topology with 8 nodes?

A. 8 B. 16 C. 28 D. 56

Answer: C. **Why:** $n(n-1)/2 = 8 \times 7/2 = 28$.

Q2. Which topology's total failure risk is concentrated entirely in one central device?

A. Bus B. Ring C. Star D. Mesh

Answer: C. **Why:** Star depends entirely on the central hub; hub failure disables the whole network.

Q3. Packet switching is preferred over circuit switching mainly because it offers:

A. Fixed delay B. Guaranteed in-order delivery C. Better bandwidth utilization via dynamic sharing D. No processing overhead per packet

Answer: C. **Why:** Packet switching shares bandwidth dynamically among users instead of reserving it, unlike circuit switching. It trades this for possible variable delay/out-of-order delivery.

Q4. X.25 maps its Data Link layer function to which protocol?

A. PLP B. LAPB C. X.21 D. TCP

Answer: B. **Why:** X.25 layering: Physical=X.21, Data Link=LAPB, Network=PLP.

Q5. Which transmission medium is most vulnerable to being blocked by a building between sender and receiver?

A. Radio waves B. Microwaves C. Infrared and Microwaves (both, being line-of-sight) D. None, all penetrate walls

Answer: C. **Why:** Both microwaves and infrared require line-of-sight and cannot penetrate solid obstacles; radio waves can.

Q6. Encryption of data for secure transmission is a standard responsibility of which OSI layer?

A. Application B. Presentation C. Session D. Transport

Answer: B. **Why:** Presentation layer handles translation, encryption/decryption, and compression — a frequently tested trap since students assume "Application layer" handles security.

Q7. ARPANET's shift that is considered the true technical birth of "the Internet" was:

A. First message sent between UCLA and SRI (1969) B. Adoption of TCP/IP in 1983 C. Launch of NSFNET in 1986 D. Introduction of WWW in 1991

Answer: B. Why: TCP/IP standardization in 1983 enabled internetworking of heterogeneous networks — the defining feature of "the Internet" as a network of networks.

Q8. A network spanning an entire city, owned by an ISP/government consortium and typically using fiber/microwave links, is classified as:

A. LAN B. MAN C. WAN D. PAN

Answer: B. Why: MAN covers ~5–50 km (city-scale) and is typically owned by a consortium/ISP, unlike a privately-owned LAN.

Q9. [Trap] A network engineer says "packet switching guarantees in-order packet delivery just like circuit switching." This is:

A. True, both guarantee order B. False — packets may take different paths and arrive out of order, requiring reassembly C. True only for the Internet D. False, because circuit switching also allows out-of-order delivery

Answer: B. Why: Only circuit switching guarantees order (fixed path); packet switching may deliver packets out of order since each packet is routed independently.

Q10. Which layer of the OSI model is responsible for adding logical (IP) addressing and routing decisions?

A. Data Link B. Network C. Transport D. Physical

Answer: B. Why: The Network layer adds logical addressing and performs routing for end-to-end delivery across multiple networks; Data Link only adds physical (MAC) addressing for node-to-node hops.

2. Data Communication Fundamentals

2.1 Analog vs Digital Signals

Feature	Analog	Digital
Nature	Continuous, infinite values	Discrete, fixed levels
Waveform	Sine wave	Square wave
Noise immunity	Low — degrades with distance	High — regenerable
Example	Voice, thermometer	Computers, CD/DVD, USB

2.2 Core Formulas (High-Yield)

Period (T) = $1 / \text{Frequency (f)}$ [T in seconds, f in Hz — inversely related]
 Bit Rate = Baud Rate $\times$ n [n = bits encoded per signal symbol]
 Throughput = actual bits successfully delivered / time taken ($\leq$ Bandwidth, always)
 Transmission Delay = Packet Size (bits) / Bandwidth (bps)
 Propagation Delay = Distance / Propagation Speed
 Total Latency = Transmission Delay + Propagation Delay (+ Queuing + Processing delay)

- Bandwidth** = theoretical max capacity of the link (Hz analog / bps digital). **Throughput** = actual delivered rate — always $\leq$ bandwidth (reduced by noise/congestion/errors).
- TRAP:** Bandwidth and Throughput are NOT the same — a question giving "link bandwidth 10 Mbps but average delivered X Mbps" is testing this gap.

Worked Example

Link bandwidth = 10 Mbps. Transmits 12,000 frames/min, 10,000 bits/frame.
 Frames/sec = $12000/60 = 200$. Throughput = $200 \times 10,000 = 2,000,000$ bps = **2 Mbps**.
 Efficiency = Throughput/Bandwidth = $2/10 = 20\%$.

2.3 Encoding Schemes — Bits per Symbol (n)

Scheme	n (bits/symbol)	Note
Manchester	0.5	Least efficient; 2 transitions/bit but self-synchronizing; legacy 10 Mbps Ethernet, still used in RFID/NFC
NRZ	1	1 pulse = 1 bit; simple but loses sync on long 0/1 runs; used in USB/UART, not long-distance
16-QAM	4	Combines amplitude + phase; old WiFi 4
4096-QAM	12	Max efficiency; needs very clean signal (short range); WiFi 7 / 5G

- TRAP:** Higher n (bits/symbol) = more efficient BUT more vulnerable to noise (needs cleaner channel).

2.4 Channel Capacity Theorems

Nyquist (NOISELESS channel): BitRate = $2 \times B \times \log_2(L)$ [B=Bandwidth(Hz), L=signal levels]
 Shannon (NOISY channel): C = $B \times \log_2(1 + \text{SNR})$ [SNR = Signal Power / Noise Power]

Feature	Nyquist	Shannon
Channel	Noiseless (ideal)	Noisy (real-world)
Depends on	Bandwidth, signal levels (L)	Bandwidth, SNR
Use	Theoretical upper bound, limited practical use	Universal real-world limit

- TRAP:** Nyquist alone is used to find max bit rate ONLY when a channel is stated as noiseless; if noise/SNR is given, use Shannon. In practice, the **actual channel capacity is the minimum of the two** if both are computed for the same channel.
- Increasing signal levels (L) increases Nyquist bit rate, but real noisy channels can't have unlimited L (error probability rises) — this is why Shannon's limit governs real channels.

2.5 Encoding / Modulation Types

- QAM** = ASK + PSK combined (varies amplitude AND phase together) → higher data rate per symbol.
- Digital-to-Digital:** Line coding (NRZ, Manchester) → Block coding (mB/nB, adds redundancy bits for sync/error-detection, $n > m$) → Scrambling (avoids long 0-runs without adding bits, keeps clock sync).
- Analog-to-Digital (PCM):** Sampling → Quantization → Encoding (binary).
- Base64:** 3 bytes (24 bits) → four 6-bit chunks → 4 printable characters (used to send binary data over text protocols like email/JSON).
- Fixed-length coding:** ASCII/EBDIC/Unicode (every symbol = same # bits). **Variable-length coding:** Huffman (compression; frequent symbols get fewer bits).

Questions

Q1. A signal has a frequency of 500 Hz. What is its period?

A. 500 s B. 2 ms C. 0.5 ms D. 5 ms

Answer: B. **Why:** $T = 1/f = 1/500 = 0.002$ s = 2 ms.

Q2. A modulation scheme encodes 4 bits per signal symbol using a baud rate of 2400 baud. Find the bit rate.

A. 600 bps B. 2400 bps C. 4800 bps D. 9600 bps

Answer: D. Why: Bit Rate = Baud Rate $\times$ n = 2400 $\times$ 4 = 9600 bps.

Q3. [Trap] Bandwidth of a link is 100 Mbps but actual measured data delivery is 40 Mbps. This 40 Mbps figure is the link's:

A. Bandwidth B. Baud rate C. Throughput D. Channel capacity

Answer: C. Why: Throughput is the actually achieved delivery rate; it is always $\leq$ bandwidth due to noise, congestion, or overhead.

Q4. A noiseless channel with bandwidth 3000 Hz uses 2 signal levels. Maximum bit rate (Nyquist) is:

A. 3000 bps B. 6000 bps C. 1500 bps D. 9000 bps

Answer: B. Why: BitRate = $2 \times B \times \log_2(L) = 2 \times 3000 \times \log_2(2) = 6000 \times 1 = 6000$ bps.

Q5. A channel has bandwidth 3 kHz and SNR = 7. Its Shannon capacity is approximately:

A. 3000 bps B. 6000 bps C. 9000 bps D. 12000 bps

Answer: C. Why: $C = B \times \log_2(1 + \text{SNR}) = 3000 \times \log_2(8) = 3000 \times 3 = 9000$ bps.

Q6. Which statement correctly distinguishes Nyquist from Shannon's theorem?

A. Nyquist applies to noisy channels, Shannon to noiseless B. Nyquist depends on SNR, Shannon on signal levels C. Nyquist gives the limit for noiseless channels using signal levels; Shannon gives the limit for noisy channels using SNR D. They give identical formulas

Answer: C. Why: Nyquist: BitRate = $2B \cdot \log_2(L)$ for a noiseless channel; Shannon: $C = B \cdot \log_2(1 + \text{SNR})$ accounts for real-world noise.

Q7. Manchester encoding is described as "least efficient" ($n = 0.5$) mainly because:

A. It cannot detect errors B. It requires 2 signal transitions to encode 1 bit, trading bandwidth for guaranteed synchronization C. It uses more signal levels than QAM D. It is only used for analog signals

Answer: B. Why: Manchester sacrifices bandwidth efficiency (2 transitions/bit) specifically to guarantee the receiver stays synchronized (signal never "flat-lines").

Q8. Base64 encoding converts every 3 bytes of binary data into how many output characters?

A. 2 B. 3 C. 4 D. 6

Answer: C. Why: 24 bits $\rightarrow$ four 6-bit groups $\rightarrow$ 4 printable characters (the "24-to-32-bit" rule).

3. Data Link Layer — Multiple Access Control

3.1 DLL Sublayers & Access Categories

- DLL splits into **LLC** (top: flow/error control, part of framing) and **MAC** (bottom: access method, e.g. CSMA/CD for Ethernet). Framing is shared by both.
- Random Access (contention)**: no station has priority; stations compete → collisions possible. (ALOHA, CSMA, CSMA/CD, CSMA/CA)
- Controlled Access**: stations coordinate; a station transmits only when authorized. (Reservation, Polling, Token Passing)
- Channelization**: bandwidth split among stations by frequency/time/code. (FDMA, TDMA, CDMA)

3.2 ALOHA

Propagation delay T_p = Distance / Propagation speed
 Transmission delay T_t ($=T_{fr}$) = Frame length / Bandwidth
 Pure ALOHA vulnerable time = $2 \times T_{fr}$
 Slotted ALOHA vulnerable time = $1 \times T_{fr}$ (half of pure ALOHA)
 Pure ALOHA max throughput $\sim 18.4\%$ (at $G = 0.5$)
 Slotted ALOHA max throughput $\sim 36.8\%$ (at $G = 1$)

- Pure ALOHA**: station sends whenever it has data; collision if any other station transmits within $\pm T_{fr}$ of it → vulnerable window = $2 \times T_{fr}$.
- Slotted ALOHA**: time divided into slots of size T_{fr} ; stations may only start at slot boundaries → collision window shrinks to just T_{fr} (halves vulnerability, doubles max throughput vs pure ALOHA).
- On collision: station waits **random back-off time T_B** , retransmits; gives up after **K_{max}** attempts.
- TRAP**: Slotted ALOHA's efficiency is exactly **double** pure ALOHA's — a favorite numerical/conceptual GATE trap.

3.3 CSMA and Persistence Strategies

- CSMA** = "sense before transmit." Reduces but does NOT eliminate collisions (propagation delay means a station may sense idle before another's signal arrives).
- Vulnerable time (CSMA) = T_p** (propagation time) — much smaller than ALOHA's vulnerable window since carrier sensing is used.
 | Method | Behavior when line found IDLE | Behavior when BUSY | Collision Risk |
 |---|---|---|---|
 | 1-persistent | Sends immediately (prob=1) | Waits, keeps sensing | Highest |
 | Non-persistent | Sends immediately | Waits **random** time, senses again | Lowest, but wastes idle time (low efficiency) |
 | p-persistent (slotted) | Sends with probability p ; else waits 1 slot & re-checks | Backs off as if collided | Balanced (best of both) |

3.4 CSMA/CD (Wired Ethernet)

Minimum frame transmission time: $T_{fr} \geq 2 \times T_p$

- A station must still be transmitting when a collision signal could return, so it can detect the collision and abort. Worst case: two max-distance-apart stations — signal takes T_p to reach the far end, and the collision effect takes another T_p to come back ⇒ **$T_{fr} \geq 2T_p$** .
- Channel energy levels: **Zero** (idle) / **Normal** (one station transmitting) / **Abnormal** (collision, $\sim 2 \times$ normal energy).
- TRAP**: This $2T_p$ rule is *why* Ethernet enforces a **minimum frame size** (64 bytes) — a frame shorter than this on a given cable length+speed cannot guarantee collision detection before it finishes sending.

3.5 CSMA/CA (Wireless — Wi-Fi)

- Collisions can't be *detected* in wireless (received signal energy too weak to distinguish from a collision) → must be *avoided*.
- Three mechanisms:
 - IFS (Interframe Space)**: wait even if channel is idle, to let far signals arrive.
 - Contention Window**: random back-off slots; doubles after each collision (**binary exponential back-off**); if channel goes busy mid-wait, timer **pauses** (not restarts) — gives priority to the longest-waiting station.
 - ACK**: explicit acknowledgment required since collisions can't be sensed directly.
- TRAP**: CSMA/CD = collision **detection** (wired); CSMA/CA = collision **avoidance** (wireless) — do not swap these.

3.6 Controlled Access

Method	Mechanism
Reservation	Time divided into Reservation Interval (M slots, 1 per station, mark intent to send) + Data period (only reserved stations send, in order).
Polling	Primary/secondary hierarchy. Select : primary → secondary ("I have data for you," waits for ready-ACK). Poll : primary asks each secondary in turn ("anything to send?" → NAK if nothing, data frame if something).
Token Passing	Logical ring; only the station holding the token may transmit; token passed to successor after use; token-holding time is capped to prevent monopolization; lost token must be regenerated.

3.7 Channelization

Scheme	Divides bandwidth by	Key idea
FDMA	Frequency	Each station gets a permanent frequency band (all the time).
TDMA	Time	Each station gets the full bandwidth but only during its time slot.
CDMA	Code (not time/frequency)	All stations transmit simultaneously over the full bandwidth, separated by unique orthogonal codes (like different "languages" in one room).

- TRAP**: CDMA is the odd one out — no time-sharing (unlike TDMA) and no frequency partition (unlike FDMA); separation is purely by code.

Questions

Q1. In Pure ALOHA, if the frame transmission time is T_{fr} , the vulnerable time during which a collision can occur is:

- A. T_{fr} B. $2 \times T_{fr}$ C. $0.5 \times T_{fr}$ D. $4 \times T_{fr}$

Answer: B. **Why:** A frame starting up to T_{fr} before or after another's frame will overlap it, giving a vulnerable window of $2 \times T_{fr}$.

Q2. [Numerical] Maximum throughput of Slotted ALOHA compared to Pure ALOHA is:

- A. Same B. Half C. Double ($\sim 36.8\%$ vs $\sim 18.4\%$) D. Four times

Answer: C. **Why:** Slotted ALOHA's max throughput ($1/e \approx 36.8\%$) is exactly double Pure ALOHA's ($1/2e \approx 18.4\%$) because slotting halves the vulnerable window.

Q3. [Trap] A student claims "CSMA guarantees no collisions since stations sense the medium before transmitting." This is:

- A. True B. False — propagation delay still allows two stations to sense idle simultaneously and collide C. True only for wireless D. True only for slotted CSMA

Answer: B. **Why:** CSMA reduces collision probability but cannot eliminate it, because a station's signal takes finite time (T_p) to reach other stations, who may sense "idle" in that window.

Q4. Why does CSMA/CD require a minimum frame transmission time of $2 \times T_p$?

- A. To reduce bandwidth usage B. So the sender is still transmitting when a collision signal could return from the farthest station C. To allow encryption of the frame D. To synchronize clocks

Answer: B. **Why:** Worst case, a collision at the far end takes T_p to occur and another T_p for its effect to propagate back — the sender must still be transmitting ($T_{fr} \geq 2T_p$) to detect it and abort.

Q5. Which multiple access method is specifically designed because collisions cannot be reliably *detected* (only avoided)?

- A. CSMA/CD B. CSMA/CA C. Pure ALOHA D. TDMA

Answer: B. **Why:** In wireless networks, received signal energy is too weak for collision detection, so CSMA/CA avoids collisions via IFS, contention window, and ACKs instead.

Q6. In IEEE 802 controlled access via Polling, a secondary station with no data to send responds to a poll with:

- A. SEL frame B. ACK frame C. NAK frame D. Token frame

Answer: C. **Why:** The primary "polls" each secondary; a secondary with nothing to send replies NAK, prompting the primary to poll the next one.

Q7. Which channelization method allows all stations to transmit simultaneously over the entire bandwidth, distinguished only by unique codes?

- A. FDMA B. TDMA C. CDMA D. Polling

Answer: C. **Why:** CDMA uses orthogonal codes for simultaneous full-bandwidth transmission — unlike FDMA (frequency split) or TDMA (time split).

Q8. [Trap] In CSMA/CA's contention window, if a station's back-off timer is running and the channel becomes busy, the station:

- A. Restarts the timer from the maximum value B. Pauses the timer and resumes when the channel is idle again C. Immediately transmits anyway D. Discards the frame

Answer: B. **Why:** The timer pauses (doesn't reset) on a busy channel — this preserves relative priority for the station that has already waited longest.

4. Data Link Layer — Flow Control (ARQ Protocols)

(Highest-yield numerical topic in CN — sliding window efficiency appears almost every year.)

4.1 Flow Control vs Error Control

- **Flow control:** prevents overwhelming a slow receiver (buffer-based feedback to sender).
- **Error control (ARQ = Automatic Repeat Request):** detects lost/corrupted/duplicate frames and triggers retransmission.

4.2 Core Delay & Efficiency Formulas

```
Tt (transmission delay) = Frame size / Bandwidth
Tp (propagation delay) = Distance / Propagation speed
RTT (Round Trip Time) = 2 x Tp (ACK size/time treated as negligible)
Total cycle time (Stop-and-Wait) = Tt + 2 x Tp
a = Tp / Tt

Efficiency (Stop-and-Wait): eta = 1 / (1 + 2a)
Throughput = eta x Bandwidth

Optimum window size to reach 100% efficiency: W = 1 + 2a
```

- If $W \geq (1+2a)$ the link is fully utilized ($\eta=1$); if $W < (1+2a)$, $\eta = W / (1+2a)$.

4.3 Stop-and-Wait ARQ

- Sender sends **1 frame**, waits for ACK before sending next → very simple, but wastes the "pipe" on high bandwidth-delay-product links.
- Adds: frame **sequence numbers** (just 0/1 — 1 bit is enough since only 1 frame is ever outstanding) to detect duplicates/lost frames; **timer** for retransmission on ACK timeout; corrupted frames are **silently discarded** (receiver "silence" is itself the error signal).
- **Bandwidth-delay product** = Bandwidth × RTT = amount of data (bits) that can be "in flight." Stop-and-Wait cannot fill this pipe on long-fat networks (large a) — hence poor efficiency for high a.

4.4 Go-Back-N ARQ (GBN)

- Sender may have **multiple frames outstanding** (sliding window) without waiting for each ACK individually → fixes Stop-and-Wait's efficiency problem.
- **Receiver window size = 1** — only accepts frames in exact order; an out-of-order/damaged frame is discarded (receiver stays silent), and ALL subsequent frames already sent get discarded by receiver too.
- **Cumulative ACK:** one ACK confirms all frames up to that number.
- On timeout for the oldest outstanding frame, sender resends **all outstanding frames from that point on** (hence "Go-Back-N" — wasteful on noisy links).
- Single timer suffices (only for the oldest unacked frame — it always expires first).

4.5 Selective Repeat ARQ (SR)

- Fixes GBN's waste: **only the specific damaged/lost frame is retransmitted**, not everything after it.
- Receiver **buffers out-of-order frames** and delivers to network layer only once a contiguous run is complete.
- Needs **individual timer per frame** and can send **NAK** for a specific damaged frame.
- Higher receiver buffer/complexity cost than GBN, but far more bandwidth-efficient on noisy/lossy links.

4.6 Sequence-Number & Window-Size Rules (m = bits used for seq. numbers, so 2^m distinct numbers)

Protocol	Sender window max	Receiver window
Stop-and-Wait	1	1
Go-Back-N	$2^m - 1$	1
Selective Repeat	2^{m-1}	2^{m-1}

- **TRAP:** Selective Repeat's window is capped at **half** of Go-Back-N's (2^{m-1} vs $2^m - 1$). Reason: if SR's window were larger, the receiver could mistake a retransmitted *old* frame for a *new* one (ambiguity in distinguishing old vs new sequence numbers when both windows can hold out-of-order frames).
- Efficiency for windowed protocols (GBN/SR) with window size N: $\eta = N/(1+2a)$ if $N < (1+2a)$, else $\eta = 1$.

4.7 Comparison

Feature	Stop-and-Wait	Go-Back-N	Selective Repeat
Frames in flight	1	Multiple (window)	Multiple (window)
Receiver window	1	1	> 1 (up to 2^{m-1})
On error	Resend that 1 frame	Resend ALL from error onward	Resend ONLY the erroneous frame
Efficiency (large a)	Poor	Good	Best (esp. on noisy links)
Complexity/Buffer	Lowest	Low	Highest (per-frame timers + buffer)

Questions

Q1. [Numerical] $T_t = 1$ ms, $T_p = 4$ ms. What is the efficiency of Stop-and-Wait ARQ?
A. 20% B. 11.1% C. 25% D. 50%

Answer: B. Why: $a = T_p/T_t = 4$; $\eta = 1/(1+2a) = 1/(1+8) = 1/9 \approx 11.1\%$.

Q2. [Numerical] For the link in Q1, what window size would achieve 100% utilization under Go-Back-N?

A. 4 B. 8 C. 9 D. 5

Answer: C. Why: Optimum $W = 1 + 2a = 1 + 8 = 9$.

Q3. A 3-bit sequence number field ($m=3$) is used. What is the maximum sender window size under Go-Back-N vs Selective Repeat respectively?

A. 8 and 4 B. 7 and 4 C. 8 and 8 D. 7 and 8

Answer: B. Why: GBN max = $2^m - 1 = 7$; Selective Repeat max = $2^{m-1} = 4$.

Q4. [Trap] Why can't Selective Repeat use the same window size as Go-Back-N ($2^m - 1$)?

A. It would need more timers than allowed B. The receiver could then misinterpret a retransmitted old frame as a new one due to sequence number ambiguity C. It would violate flow control rules D. Selective Repeat doesn't use sequence numbers at all

Answer: B. Why: With overlapping large windows on both ends, an old retransmitted frame's sequence number could fall within the receiver's new expected range, causing ambiguity — hence the window is capped at half.

Q5. In Go-Back-N, if frames 4, 5, 6, 7 are sent and frame 4's timer expires (others undelivered/unacknowledged), the sender resends:

A. Only frame 4 B. Frames 4, 5, 6, 7 C. Only frame 7 D. Nothing, it waits for a NAK

Answer: B. Why: Go-Back-N resends the timed-out frame AND all subsequent outstanding frames — the defining (and wasteful) behavior that Selective Repeat improves upon.

Q6. Which ARQ protocol requires the receiver to maintain a reordering buffer for out-of-order frames?

A. Stop-and-Wait B. Go-Back-N C. Selective Repeat D. All of the above

Answer: C. Why: Only Selective Repeat has a receiver window > 1 , so it must buffer out-of-order frames until a contiguous sequence can be delivered upward.

Q7. [Conceptual] Why does Stop-and-Wait ARQ need only 1 bit for sequence numbers?

A. Because errors never occur B. Because at most one frame is ever outstanding — only "current" and "next" need distinguishing (0/1 suffices) C. Because ACKs are numbered instead D. Because the window size is $2^m - 1$

Answer: B. Why: Since the sender waits for each ACK before sending the next frame, only two states (current frame, next frame) ever need distinguishing — hence sequence numbers cycle 0,1,0,1...

5. Data Link Layer — Error Control (Hamming, CRC, Checksum)

5.1 Error Types & Redundancy

- **Single-bit error**: exactly 1 bit flipped. **Burst error**: ≥ 2 bits flipped; length = span from first to last corrupted bit (bits in between need not all be wrong).
- **Error Detection** (know *that* an error occurred) is simpler than **Error Correction** (know *where*, to fix it).
- **FEC (Forward Error Correction)**: receiver reconstructs data itself using redundant bits (no retransmission needed) — used when retransmission is costly/impossible (e.g., live streaming).
- **Retransmission (ARQ)**: receiver asks sender to resend — see Ch.4.

5.2 Block Coding & Hamming Distance

- Datword k bits $\rightarrow$ Codeword n bits ($n=k+r$, r =redundant bits). Of 2^n possible codewords, only 2^k are valid; **$2^n - 2^k$ are invalid/unused** — an error is detected if the received word lands on an invalid codeword.
- **Hamming distance $d(x,y)$** = number of differing bit positions = $\text{popcount}(x \oplus y)$.
- **$d_{\min}$** = minimum Hamming distance over all pairs of valid codewords.

5.3 The Governing Rules (Very High-Yield)

To DETECT up to s errors: $d_{\min} \geq s + 1$
 To CORRECT up to t errors: $d_{\min} \geq 2t + 1$
 To DETECT s errors AND CORRECT t errors ($s \geq t$): $d_{\min} \geq s + t + 1$

- **TRAP**: " $d_{\min} = 3$ " is commonly misread. $d_{\min}=3$ allows you to EITHER detect up to 2 errors OR correct up to 1 error — not both simultaneously at those magnitudes.
- Coding scheme notation: **$C(n, k)$** with a stated $d_{\min}$, e.g. $C(7,4)$, $d_{\min}=3$.

5.4 Parity Check Codes

Code	Redundant bits	Detects	$d_{\min}$
Simple parity	1 bit ($n=k+1$)	Odd number of bit errors only	2
2D (row+column) parity	1 row per row + 1 column per col	Up to 3 errors anywhere in the table; 4-bit errors may go undetected	—

- **TRAP**: Simple parity CANNOT detect an even number of flipped bits (they cancel out and parity looks fine).

5.5 Hamming Code (single-bit error correction)

r redundant bits $\Rightarrow n = 2^r - 1$, $k = n - r = 2^r - r - 1$
 Parity bits placed at positions that are powers of 2: 1, 2, 4, 8, ...

- Classic exam case: **$r=3 \rightarrow C(7,4)$, $d_{\min}=3$** $\rightarrow$ corrects any single-bit error.
- Parity bit **P1** checks positions with bit-0 set (1,3,5,7); **P2** checks positions with bit-1 set (2,3,6,7); **P4** checks positions with bit-2 set (4,5,6,7). At the receiver, the syndrome (P1P2P4 read as a binary number) directly gives the **position** of the error bit — this "self-locating" property is the key exam insight.
- **TRAP**: Hamming code positions are 1-indexed and use POWERS OF 2 for the parity/check bits, not the first r positions.

5.6 Checksum

- Sender adds up all data words $\rightarrow$ sends the **sum** (or its complement) along with the data.
- Receiver re-adds everything (including the checksum); if using ordinary sum, compare to sent value; if using **1's-complement sum**, the total should be all 1s (or, equivalently, adding the complement makes the total = 0).
- Uses **1's complement arithmetic** (end-around carry) — a wraparound-add scheme, not 2's complement.
- **TRAP**: Checksum is a *weak* error-detection method (used at Transport/IP layers) — it can miss certain compensating errors that CRC would catch. CRC is stronger and used at the Data Link layer/hardware level.

5.7 Cyclic Redundancy Check (CRC)

- **Cyclic code property**: any cyclic shift of a valid codeword is also a valid codeword.
- Procedure: append $(n-k)$ zeros to the k -bit dataword $\rightarrow$ divide by a fixed generator (degree $n-k$, so $n-k+1$ bits) using **modulo-2 (XOR) division** $\rightarrow$ the remainder ($n-k$ bits) replaces the appended zeros to form the final n -bit codeword.
- Receiver divides the received codeword by the same generator; **remainder = 0 $\rightarrow$ no error detected**; nonzero remainder ("syndrome") $\rightarrow$ error.
- Polynomial view: bit string $\leftrightarrow$ polynomial (bit position = power of x); a 7-bit string has degree ≤ 6 . Left-shift by n bits $\equiv$ multiply by x^n .

Worked CRC Example (classic)

Dataword = **1001** ($k=4$), Generator = **1011** ($= x^3+x+1$, so $n-k=3$).
 Augmented dataword = 1001 000 (append 3 zeros) $= x^6+x^3$.
 Divide (mod-2) by $x^3+x+1 \rightarrow$ **remainder = 110**.
 Final codeword transmitted = **1001 110**.
 (Receiver divides 1001110 by 1011; remainder 000 $\Rightarrow$ accepted as error-free.)

Questions

Q1. [Numerical] Find the Hamming distance between 10110 and 11100.
 A. 1 B. 2 C. 3 D. 4

Answer: B. **Why:** XOR: $10110 \oplus 11100 = 01010 \rightarrow$ two 1s $\rightarrow d=2$.

Q2. A coding scheme has $d_{\min} = 5$. What is the maximum number of errors it can guarantee to CORRECT?

A. 5 B. 4 C. 2 D. 3

Answer: C. Why: $d_{\min} \geq 2t+1 \Rightarrow 5 \geq 2t+1 \Rightarrow t \leq 2 \rightarrow$ corrects up to 2 errors.

Q3. [Trap] The same code from Q2 ($d_{\min}=5$) — what is the max number of errors it can guarantee to DETECT (not correct)?

A. 2 B. 3 C. 4 D. 5

Answer: C. Why: $d_{\min} \geq s+1 \Rightarrow 5 \geq s+1 \Rightarrow s \leq 4 \rightarrow$ detects up to 4 errors (detection capacity is always $\geq$ correction capacity for the same $d_{\min}$).

Q4. A Hamming code is designed with $r=4$ redundant bits. What are n and k ?

A. $n=15, k=11$ B. $n=16, k=12$ C. $n=15, k=10$ D. $n=14, k=10$

Answer: A. Why: $n = 2^r - 1 = 15$; $k = n - r = 15 - 4 = 11 \rightarrow C(15,11)$.

Q5. In a 7-bit Hamming code ($C(7,4)$), which bit positions hold the parity bits?

A. 1, 2, 3 B. 1, 2, 4 C. 4, 5, 6 D. 1, 4, 7

Answer: B. Why: Parity bits occupy positions that are powers of 2: 1, 2, 4 (P_1, P_2, P_4); data bits occupy the rest (3,5,6,7).

Q6. [Trap] Simple parity check fails to detect an error when:

A. Exactly 1 bit flips B. Exactly 3 bits flip C. Exactly 2 bits flip (or any even number) D. No bits flip

Answer: C. Why: Simple parity only preserves odd/even 1-count; an even number of flips restores the original parity, hiding the error — a textbook trap on "always detects errors" claims.

Q7. [Numerical] Dataword 1101 is sent with CRC generator 1011 ($n-k=3$). What must the remainder be for the receiver to accept the frame as error-free?

A. Equal to the generator B. Equal to the original dataword C. All zeros (000) D. Equal to 1011

Answer: C. Why: The receiver divides the received codeword by the same generator; a zero remainder (syndrome) means no detected error.

Q8. Which error-detection technique is generally considered stronger and hardware-implemented at the Data Link layer, compared to the simpler checksum used at Transport/IP layers?

A. Parity bit B. Checksum C. CRC D. Hamming code

Answer: C. Why: CRC (via polynomial division) catches a much wider class of error patterns than the additive checksum, which is why CRC is standard in Ethernet frames while checksum suffices for IP/TCP/UDP headers.

6. Data Link Layer — Framing & Ethernet

6.1 Framing Types

- **Fixed-size framing:** frame length itself is the delimiter (no explicit boundary markers needed).
- **Variable-size framing:** needs explicit boundary markers —
- **Character-oriented:** uses a flag byte + **byte stuffing** (insert ESC before any data byte matching the flag/ESC pattern).
- **Bit-oriented:** uses flag bit pattern **01111110** + **bit stuffing** (insert a 0 after every 5 consecutive 1s in the data, to prevent the flag pattern appearing inside data; receiver removes the stuffed 0).
- **TRAP:** Byte stuffing inserts an extra **byte** (ESC); bit stuffing inserts a single extra **bit** (0) — don't conflate the two mechanisms.

6.2 Ethernet (IEEE 802.3)

- Standardized **1983**; originally **10BASE5** (coaxial, bus topology, **Manchester encoding**); connectionless & **unreliable** (no built-in ACK — corrupted frames are silently dropped; retransmission is left to higher layers, e.g., TCP).
- Uses **48-bit MAC addresses**.

Ethernet Frame Format

| Preamble (7B) | SFD (1B) | Dest MAC (6B) | Src MAC (6B) | Length (2B) | Data (46-1500B) | CRC (4B) |

- **Preamble (7B):** alternating 0/1 pattern for bit synchronization — added at Physical layer.
- **SFD (1B) = 10101011:** trailing "11" signals start of the actual frame.
- **Data field range: 46–1500 bytes.** If actual data < 46B → padded with 0s. If > 1500B → must be fragmented across multiple frames.
- **CRC (4B):** computed over the frame, placed last since it must be calculated after the rest of the frame is assembled.
- **Minimum frame size = 64 bytes** (6+6+2+46+4 = 64, excluding preamble+SFD) — this exists specifically to satisfy the **CSMA/CD rule Tfr ≥ 2Tp** (Ch.3 §3.4), ensuring collision detection works before the frame finishes transmitting.
- Maximum length cap exists for 2 historical reasons: (1) memory was expensive (buffer size), (2) prevents one station from monopolizing the shared medium.

MAC Address Types (via LSB of the first byte)

Type	Rule	Meaning
Unicast	LSB of first byte = 0	One specific destination (source address is ALWAYS unicast)
Multicast	LSB of first byte = 1	A group of stations
Broadcast	All 48 bits = 1 (FF:FF:FF:FF:FF:FF)	Every station on the LAN

6.3 Manchester Encoding

- Each bit period is split in half: signal is High-then-Low (or vice versa) — the **mid-bit transition itself provides clock synchronization** (self-clocking).
- Used in classic 10 Mbps Ethernet. (See Ch.2 §2.3 for its $n=0.5$ bits/symbol efficiency cost.)

6.4 PPP (Point-to-Point Protocol) — quick reference

- Establishes a direct link between two nodes over serial/telephone lines; encapsulates Network-layer packets; supports authentication (**PAP, CHAP**); used in DSL/dial-up; replaced the older **SLIP**.

Questions

Q1. Why is bit stuffing needed in bit-oriented framing?

A. To compress the data B. To prevent the flag pattern 01111110 from appearing accidentally inside the data stream C. To add error correction D. To encrypt the frame

Answer: B. **Why:** A 0 is inserted after every 5 consecutive 1s in the data so the receiver never mistakes real data for the frame-delimiting flag.

Q2. [Numerical] What is the minimum valid Ethernet frame size (excluding preamble & SFD)?

A. 46 bytes B. 60 bytes C. 64 bytes D. 1500 bytes

Answer: C. **Why:** $6(\text{Dest}) + 6(\text{Src}) + 2(\text{Length}) + 46(\text{min data}) + 4(\text{CRC}) = 64$ bytes — required to satisfy the CSMA/CD 2Tp collision-detection window.

Q3. A MAC address has its first byte's LSB = 1. This address is:

A. Unicast B. Multicast C. Broadcast D. Invalid

Answer: B. **Why:** LSB=1 in the first byte marks a multicast (group) address; LSB=0 marks unicast; all-1s (FF...FF) is broadcast.

Q4. [Trap] Since Ethernet has no built-in acknowledgment mechanism, reliability (retransmission of lost data) is handled by:

A. The Physical layer B. The Ethernet CRC field automatically C. Higher layer protocols (e.g., TCP) D. The Preamble

Answer: C. **Why:** Ethernet is connectionless and unreliable by design — corrupted frames are silently dropped; any needed retransmission is the job of a higher-layer protocol like TCP.

Q5. Why must the CRC field be placed at the end of an Ethernet frame rather than the beginning?

A. Historical convention only B. Because CRC must be calculated after the rest of the frame's bits are known/assembled C. Because CRC is part of the Preamble D.

Because MAC addresses must come first for switching

Answer: B. **Why:** CRC is a checksum over the frame contents, so it can only be computed once the full frame (addresses + length + data) is ready.

Q6. Manchester encoding embeds clock synchronization by:

A. Sending a separate clock signal B. A guaranteed voltage transition at the mid-point of every bit C. Using only high voltage for all bits D. Encrypting the bit stream

Answer: B. **Why:** The mandatory mid-bit transition means the signal never "flat-lines," letting the receiver continuously resynchronize — at the cost of needing 2 transitions per bit ($n=0.5$ efficiency).

7. Network Layer — IPv4 Protocol

7.1 Network Layer Role

- **Services:** Logical addressing, Routing (best path via routers), Packetizing (encapsulation with header), Error control (header checksum + ICMP), Congestion control.
- **TRAP:** Network layer does **not** provide flow control directly (that's DLL/Transport's job) — a frequently misattributed function.
- **IPv4** = unreliable, connectionless, **best-effort**, datagram (packet-switched) protocol. "Best-effort" $\Rightarrow$ packets may be corrupted, lost, duplicated, delayed, or arrive **out of order**. Reliability must be added by a higher layer (TCP).

7.2 IPv4 Datagram Header

Header length = 20 to 60 bytes (variable, in units of 4-byte words)
 HLEN field (4 bits) stores header length / 4 $\Rightarrow$ actual header length = HLEN $\times$ 4
 Range of HLEN field value = [5, 15] (5 $\times$ 4=20 min, 15 $\times$ 4=60 max)
 Total Length field (16 bits) = header + data, max = 65535 bytes
 Data length = Total Length - (HLEN $\times$ 4)

Field	Size	Purpose
Version	4 bits	IPv4 = value 4
HLEN	4 bits	Header length $\div$ 4 (see above)
Type of Service/Precedence	8 bits	QoS priority; low-precedence datagrams dropped first under congestion
Total Length	16 bits	Header + data, in bytes
Identification	16 bits	Unique per original datagram; copied unchanged to every fragment for reassembly
Flags	3 bits	bit0 reserved, D (Don't Fragment) , M (More Fragments)
Fragmentation Offset	13 bits	Position of fragment's data in original datagram, in units of 8 bytes
TTL	8 bits	Hop limit
Protocol	8 bits	Which upper-layer protocol (TCP/UDP/...) gets the payload
Header Checksum	16 bits	Covers header only , not payload
Source/Dest Address	32 bits each	IPv4 addresses
Options	0–40 bytes	Optional; testing/debugging/routing control

- Minimum total datagram length = 20 bytes (header only, no data).

7.3 Fragmentation & MTU

- **MTU** = max payload a link-layer frame can carry (e.g., typical LAN Ethernet MTU = 1500 bytes). If a datagram > next-hop MTU $\rightarrow$ **fragmented**.
- Fragmentation can occur at the **source or any intermediate router**; **only the final destination reassembles** (not intermediate routers).
- A datagram may be fragmented multiple times across different MTU links en route.
- **D bit = 1** $\Rightarrow$ must NOT fragment (if too big, it's dropped + ICMP error). **M bit = 1** $\Rightarrow$ more fragments follow; **M=0** $\Rightarrow$ last (or only) fragment.
- **Fragmentation offset** = (starting byte position of this fragment's data in original datagram) / 8.

Worked Fragmentation Example

Original datagram data = 4000 bytes (numbered 0–3999); MTU allows 1400-byte data chunks.

Fragment 1: bytes 0–1399 $\rightarrow$ offset = 0/8 = **0**.

Fragment 2: bytes 1400–2799 $\rightarrow$ offset = 1400/8 = **175**.

Fragment 3: bytes 2800–3999 $\rightarrow$ offset = 2800/8 = **350**.

(Each fragment's own Total Length = its data size + its own header size; all fragments share the same Identification value; only the last fragment has M=0.)

7.4 TTL, Protocol & Checksum Fields

- **TTL (Time To Live):** decremented by 1 at every router hop; **reaches 0 $\rightarrow$ datagram discarded** (router sends ICMP Time Exceeded). Prevents infinite routing loops. Setting TTL=1 intentionally confines a packet to the local network (used by `traceroute`).
- **Protocol field:** tells the destination which upper-layer protocol (TCP/UDP/ICMP/IGMP) should receive the payload.
- **Header Checksum:** covers **ONLY the header**, not the payload — because (a) TTL (and other header fields) change at every hop, forcing recalculation each time, and (b) upper layers (TCP/UDP) already checksum the full payload, so re-checking payload at IP level would be redundant and slow.

7.5 IP Options (variable part, up to 40 bytes, optional)

Option	Function
End of Option	Padding/terminator
Record Route	Logs up to 9 router addresses traversed (debugging)
Strict Source Route	Sender dictates the EXACT route; deviation $\Rightarrow$ datagram discarded
Loose Source Route	Must visit listed routers, but additional routers are also allowed
Timestamp	Records processing time at each router (clock sync not guaranteed)

7.6 Companion Network-Layer Protocols

Protocol	Maps	Direction
ARP	IP address → MAC (link-layer) address	Forward — "I know your IP, what's your MAC?" (broadcast request, unicast reply)
RARP	MAC address → IP address	Reverse — used by diskless machines at boot to learn their own IP
ICMP	—	Error reporting + diagnostic queries for IP (IP itself has no error-reporting)
IGMP	—	Manages multicast group membership (one-to-many delivery)

- **TRAP:** ARP and RARP are exact **inverses** of each other — ARP: IP→MAC; RARP: MAC→IP. Easy to swap under exam pressure.
- **ICMP reports errors, it does NOT correct them** — correction/retransmission is TCP's job.
- **ICMP error messages are suppressed for:** datagrams carrying another ICMP error message, non-first fragments, multicast destinations, and special addresses (127.0.0.0, 0.0.0.0) — prevents infinite error-loops.
- **ICMP message types:** Error-reporting (Destination Unreachable, Time Exceeded, Parameter Problem, Redirect) vs Query pairs (Echo Request/Reply = ping ; Router Solicitation/Advertisement; Address-Mask Request/Reply; Timestamp Request/Reply for RTT/clock sync).

Questions

Q1. [Numerical] An IPv4 header's HLEN field contains binary value 0110. What is the actual header length?

A. 6 bytes B. 20 bytes C. 24 bytes D. 30 bytes

Answer: C. **Why:** 0110 = 6; header length = $6 \times 4 = 24$ bytes.

Q2. [Numerical] A datagram has Total Length = 1500 bytes and HLEN = 5. How much actual data (payload) does it carry?

A. 1500 bytes B. 1480 bytes C. 1475 bytes D. 1420 bytes

Answer: B. **Why:** Header = $5 \times 4 = 20$ bytes; Data = Total Length – Header = $1500 - 20 = 1480$ bytes.

Q3. [Numerical] A 3000-byte payload must be fragmented for a link with MTU allowing 1000 bytes of data per fragment (multiple of 8). What is the fragmentation offset of the 3rd fragment?

A. 100 B. 200 C. 250 D. 1000

Answer: C. **Why:** 3rd fragment starts at byte 2000; offset = $2000/8 = 250$.

Q4. [Trap] If the "Don't Fragment" (D) bit is set to 1 and the datagram exceeds the outgoing link's MTU, what happens?

A. It is fragmented anyway B. It is dropped, and an ICMP error is sent to the sender C. It is queued until the link is upgraded D. The M bit is automatically set to 1

Answer: B. **Why:** D=1 forbids fragmentation; a router that cannot forward the oversized datagram intact must discard it and report the problem via ICMP.

Q5. Why is the IPv4 header checksum recalculated at every router hop?

A. Because the payload changes at each hop B. Because fields like TTL change at every hop, altering the header C. Because IP addresses change at each hop D. It is never recalculated

Answer: B. **Why:** TTL decrements at each hop, changing header content, so the header checksum (which covers header only) must be recomputed each time.

Q6. Which protocol allows a diskless machine to discover its own IP address using only its known MAC address?

A. ARP B. RARP C. ICMP D. IGMP

Answer: B. **Why:** RARP performs the reverse mapping (MAC→IP), historically used for diskless workstation booting.

Q7. Which ICMP message is generated when a router's decremented TTL field reaches zero?

A. Destination Unreachable B. Redirect C. Time Exceeded D. Parameter Problem

Answer: C. **Why:** TTL reaching 0 causes the router to discard the datagram and notify the source with a Time Exceeded message.

Q8. [Trap] Which of the following will NOT trigger an ICMP error message?

A. A normal unicast datagram whose TTL expires B. A datagram that is itself carrying an ICMP error message C. A datagram with a bad header checksum D. A datagram whose destination port is unreachable

Answer: B. **Why:** ICMP explicitly suppresses generating error messages about datagrams that already carry ICMP error messages, to prevent infinite error-reporting loops.

Q9. IGMP exists primarily to support which type of communication?

A. Unicast B. Broadcast C. Multicast D. Anycast

Answer: C. **Why:** IGMP manages group membership so routers know which hosts want a given multicast stream, enabling one-to-many delivery.

8. Network Layer — IP Addressing

(Highest-yield numerical topic in the whole CN syllabus — subnetting/CIDR/VLSM appear almost every year.)

8.1 IPv4 Basics

- IPv4 address = **32 bits** → max 2^{32} (~4.29 billion) addresses. IPv6 = 128 bits (~ 3.4×10^{38}).
- Structure: **Prefix (Network ID) + Suffix (Host ID)**. All hosts in one network share the same Network ID; different networks may reuse the same Host ID.
- Notation: Binary (32 bits, 4 octets) or Dotted-Decimal (e.g., 117.149.29.2).
- Only network-layer devices (hosts, routers) have IP addresses — **switches, hubs, repeaters do NOT** (they operate below the network layer).

8.2 Classful Addressing

Class	1st bit(s)	1st octet range	NetID bits	HostID bits	#Networks	#Hosts/network
A	0	0–127	8	24	$2^7 - 2 = 126$	$2^{24} - 2 = 16,777,214$
B	10	128–191	16	16	$2^{14} = 16,384$	$2^{16} - 2 = 65,534$
C	110	192–223	24	8	2^{21}	$2^8 - 2 = 254$
D	1110	224–239	— (multicast)	—	—	—
E	1111	240–255	— (reserved/experimental)	—	—	—

Formula: Hosts per network = $2^{(\text{host bits})} - 2$ [-2 for Network ID (all-0 host bits) and Broadcast (all-1 host bits)]
 Networks in class = $2^{(\text{netid bits used, excluding class-marker bits})}$ [minus reserved, for Class A]

- Class A loses 2 networks to reserved addresses: **0.0.0.0** (broadcast/unspecified) and **127.0.0.0/8** (loopback, e.g. 127.0.0.1 — used for local software testing) → usable range [1,126].
- TRAP:** The “-2” applies twice for different reasons — once for reserved network numbers (Class A only) and always for host addresses in every network (network address + broadcast address).

8.3 Casting Types

- Unicast:** 1-to-1. **Multicast:** 1-to-many (specific group, Class D). **Broadcast:** 1-to-all.
- Limited broadcast:** all 32 bits = 1 → **255.255.255.255** (same network only).
- Direct broadcast:** Network ID fixed + Host ID all 1s → reaches all hosts of a *specific other* network (e.g., 200.1.2.255 for network 200.1.2.0/24).

8.4 Subnetting — Why & FLSM

- Why subnet:** improves security (isolate departments), simplifies management of huge Class A/B blocks.
- Disadvantages:** loses 2 addresses per subnet (net ID + broadcast), increases routing table size, adds complexity.
- FLSM (Fixed-Length Subnet Mask):** every subnet is the SAME size (same mask, same host count) — simple but wastes addresses if actual host needs vary.
- VLSM (Variable-Length Subnet Mask):** subnets sized to actual need — no wastage, but harder to design (must allocate **largest subnets first**, so start addresses stay divisible by block size).
- Subnet mask = 32-bit number, 1s = Network+Subnet ID portion, 0s = Host ID portion. Default masks: Class A=255.0.0.0, B=255.255.0.0, C=255.255.255.0.

Worked FLSM Example — 200.1.2.0/24 split into 4 equal subnets

Need 4 subnets => borrow 2 bits => each subnet size = $256/4 = 64$ addresses => /26 mask (255.255.255.192)

Subnet 1: 200.1.2.0 - 200.1.2.63 | hosts [200.1.2.1 - 200.1.2.62] | broadcast 200.1.2.63
 Subnet 2: 200.1.2.64 - 200.1.2.127 | hosts [200.1.2.65 - 200.1.2.126] | broadcast 200.1.2.127
 Subnet 3: 200.1.2.128 - 200.1.2.191 | hosts [200.1.2.129 - 200.1.2.190] | broadcast 200.1.2.191
 Subnet 4: 200.1.2.192 - 200.1.2.255 | hosts [200.1.2.193 - 200.1.2.254] | broadcast 200.1.2.255
 (each subnet: 64 addresses, 62 usable hosts = $64-2$)

Worked VLSM Example — 14.24.74.0/24 needs subnets of 120, 60, 10 hosts

Step 1: round each requirement up to nearest (power of 2 - 2):
 120 hosts -> need 122+ addresses -> 128 ($=2^7$) -> /25 mask
 60 hosts -> need 62+ addresses -> 64 ($=2^6$) -> /26 mask
 10 hosts -> need 12+ addresses -> 16 ($=2^4$) -> /28 mask

Step 2: allocate LARGEST block first so start addresses stay aligned:
 120-block: 14.24.74.0/25 -> range .0-.127 | hosts .1-.126 | broadcast .127
 60-block: 14.24.74.128/26 -> range .128-.191 | hosts .129-.190 | broadcast .191
 10-block: 14.24.74.192/28 -> range .192-.207 | hosts .193-.206 | broadcast .207
 (Unused/reserved for future growth: .208-.255)

- TRAP:** If you allocate smallest-first in VLSM, later larger blocks may not start at an address divisible by their own size — always allocate **largest requirement first**.

8.5 CIDR (Classless Inter-Domain Routing)

- Solves classful addressing's rigidity (Class A/B too big, C too small, E wasted) — notation **a.b.c.d/n** (n = prefix/network bits).

Number of addresses in block N = $2^{(32-n)}$
 First address = (any address in block) AND (mask)
 Last address = (any address in block) OR (NOT mask)

Rules for a valid CIDR block

1. All addresses must be **contiguous**.
2. Block size must be a **power of 2**.
3. The first address must be **divisible by the block size**.

Worked Example — 167.199.170.82/27

```
/27 => mask = 255.255.255.224 => host bits = 5 => block size = 2^5 = 32 addresses
First address = address AND mask = 167.199.170.64
Last address = address OR (NOT mask) = 167.199.170.95
=> Block range: 167.199.170.64 - 167.199.170.95 (32 addresses)
```

Quick validity checks (apply the 3 rules)

- Range 100.1.2.32–100.1.2.47: size=16 ($=2^4$), 32 is divisible by 16 → **Valid CIDR block**, representation = 100.1.2.32/28.
- Range 150.10.20.64–150.10.20.127: size=64 ($=2^6$), 64 divisible by 64 → **Valid**, = 150.10.20.64/26.

8.6 Supernetting (CIDR route aggregation)

- Combines multiple contiguous same-size blocks into one larger block (reduces routing table size).
- Rules: blocks must be **contiguous**, and the **first network ID must be divisible by the combined block's size**.
- Example: 128.56.24.0/24, 128.56.25.0/24, 128.56.26.0/24, 128.56.27.0/24 → 4 contiguous /24s ($4 \times 256 = 1024 = 2^{10}$ addresses) → first ID 24 is divisible by 4 → valid supernet = **128.56.24.0/22**.
- **TRAP**: Aggregation FAILS if blocks are non-contiguous or the first ID isn't divisible by the combined size — e.g., 100.1.2.0/25 + 100.1.2.128/26 + 100.1.3.192/26 cannot form one supernet (there's a gap between 100.1.2.192 and 100.1.3.191 that isn't covered).

8.7 Special / Reserved Addresses

- **127.0.0.0/8** — Loopback (self-test, e.g. 127.0.0.1).
- **0.0.0.0** — unspecified/this-host.
- **255.255.255.255** — limited broadcast.
- **Private IP ranges (not routable on public Internet):**

```
10.0.0.0 - 10.255.255.255 (10.0.0.0/8)
172.16.0.0 - 172.31.255.255 (172.16.0.0/12)
192.168.0.0 - 192.168.255.255 (192.168.0.0/16)
```

- An ISP typically provides: **IP address, Default Gateway (DGW), Subnet Mask (SM), DNS server**.

Questions

Q1. [Numerical] How many usable host addresses are available in a Class B network?

A. 65,536 B. 65,534 C. 16,384 D. 254

Answer: B. **Why:** Class B has 16 host bits → $2^{16} - 2 = 65,534$ (subtracting network ID and broadcast addresses).

Q2. [Numerical] A network 192.168.10.0/24 is divided into 8 equal subnets. What is the new subnet mask and how many usable hosts per subnet?

A. /26, 62 hosts B. /27, 30 hosts C. /28, 14 hosts D. /27, 32 hosts

Answer: B. **Why:** 8 subnets need 3 borrowed bits → /27 (255.255.255.224); each subnet has $2^5 = 32$ addresses → $32 - 2 = 30$ usable hosts.

Q3. [Numerical] Using VLSM on 14.24.74.0/24 for subnets needing 120, 60, and 10 hosts (allocated largest-first), what is the broadcast address of the 60-host subnet?

A. 14.24.74.127 B. 14.24.74.191 C. 14.24.74.207 D. 14.24.74.255

Answer: B. **Why:** 120-host block takes /25 (.0–.127); the 60-host block starts right after at .128 and needs /26 (64 addresses) → range .128–.191 → broadcast = .191.

Q4. [Trap] Why must VLSM allocate the largest subnet requirement FIRST?

A. It doesn't matter, order is irrelevant B. So each block's starting address stays properly aligned/divisible by its own size C. To save router memory D. Because smaller blocks must always get lower addresses

Answer: B. **Why:** A CIDR/VLSM block's start address must be divisible by its block size; assigning large blocks first keeps every subsequent block's start address correctly aligned.

Q5. [Numerical] Find the number of addresses, first address, and last address for 167.199.170.82/27.

A. 16 addresses, .64–.79 B. 32 addresses, .64–.95 C. 64 addresses, .0–.63 D. 32 addresses, .82–.113

Answer: B. **Why:** /27 → 5 host bits → $2^5 = 32$ addresses; mask=255.255.255.224; First = address AND mask = .64; Last = address OR NOT(mask) = .95.

Q6. [Numerical/Trap] Is the range 100.1.2.20–100.1.2.35 a valid CIDR block?

A. Yes, it's /28 B. Yes, it's /27 C. No — the block size (16) is a power of 2, but the first address (20) is not divisible by 16 D. No, because the range isn't contiguous

Answer: C. **Why:** Size = $35 - 20 + 1 = 16 = 2^4$ (power of 2, rule 2 OK), but $20 \div 16$ is not an integer, so rule 3 (first address divisible by block size) fails — not a valid CIDR block.

Q7. Four contiguous blocks 128.56.24.0/24 through 128.56.27.0/24 are aggregated. What is the resulting supernet?

A. 128.56.24.0/23 B. 128.56.24.0/22 C. 128.56.0.0/16 D. 128.56.24.0/25

Answer: B. **Why:** $4 \times 256 = 1024 = 2^{10}$ addresses, and 24 is divisible by 4 (the block count) → valid aggregation into a /22 ($32 - 10 = 22$).

Q8. Which of these is a valid private IP address range?

A. 11.0.0.0 – 11.255.255.255 B. 172.16.0.0 – 172.31.255.255 C. 192.169.0.0 – 192.169.255.255 D. 127.0.0.0 – 127.255.255.255

Answer: B. **Why:** 172.16.0.0/12 is one of the three reserved private ranges (along with 10.0.0.0/8 and 192.168.0.0/16); 127.x is loopback, not private-use.

Q9. [Conceptual] What is the fundamental disadvantage of FLSM compared to VLSM?

A. FLSM cannot use CIDR notation B. FLSM wastes addresses when actual subnet host-count needs differ, since every subnet must be the same size C. FLSM cannot be used with Class C D. FLSM requires more routing entries than VLSM

Answer: B. **Why:** FLSM forces all subnets to the same fixed size (sized for the largest requirement), wasting addresses in smaller subnets — VLSM sizes each subnet to its actual need.

Q10. [Trap] A router receives a packet for 127.0.0.1. This address is:

A. A valid public destination on the Internet B. A loopback address used for local self-testing, never routed onto a real network C. The default gateway address D. A broadcast address

Answer: B. **Why:** The entire 127.0.0.0/8 block is reserved for loopback/local testing — it never actually leaves the host.

9. Network Layer — Routing Protocols

9.1 Routing Basics

- **Routing table:** tells a router which interface to use to forward a packet toward a destination. **Routing** = building/updating this table; **switching** = actually forwarding a packet along the chosen path.
- **Static routing table:** manually configured; impractical for large/changing networks like the Internet.
- **Dynamic routing table:** auto-updates as topology/traffic changes — required for the Internet's scale.
- **Flooding** (routing-table-free alternative): forward an incoming packet out on every interface except the one it arrived on. Guarantees shortest-path delivery and high reliability, but causes massive **duplicate packets** and traffic — needs hop-count/sequence-number limits to avoid infinite propagation.
- **Autonomous System (AS):** a group of networks/routers under one administrative domain. **Intra-domain routing** = within one AS (e.g., RIP, OSPF). **Inter-domain routing** = between ASes (uses one standard protocol, e.g., BGP).

9.2 Distance Vector Routing (DVR)

- Each router keeps a table of (destination, cost, next-hop) and shares **only destination+cost** (not next-hop) with its **immediate neighbors only** — based on **Bellman-Ford** algorithm.
- Updates: **Periodic** (e.g., every 30s) and **Triggered** (on detecting a change/link failure).
- **Count-to-Infinity / Two-Node Loop Instability:** if a link to destination X fails, a neighbor that still has stale "I can reach X" info keeps advertising it; the two nodes then loop, each believing the other has a path, slowly incrementing cost toward infinity instead of converging immediately.
- **Split Horizon:** a router never advertises a route back to the neighbor it learned that route from (since that neighbor already knows it, and it prevents feeding back stale info) — reduces but does not fully eliminate count-to-infinity.
- **TRAP:** DVR shares information with **neighbors only**, not the whole network — this is the opposite of Link State Routing (see below); a very common mix-up.

9.3 RIP (Routing Information Protocol) — a real DVR implementation

- Metric = **hop count**; same cost per hop regardless of link speed.
- **Max hop count = 15; 16 = unreachable (infinity).**
- Runs over **UDP port 520** at the application layer.
- Timers: Periodic update timer; **Expiration timer (180s)** — no update $\Rightarrow$ route marked expired; **Garbage collection timer (120s after expiry)** — route then removed.
- Fast convergence *because* of the small hop-count ceiling, but this small ceiling also limits RIP to small networks.

9.4 Link State Routing (LSR)

- Every router builds a **complete map of the entire network topology** (nodes, links, costs, up/down state) by **flooding Link State Packets (LSPs) to ALL routers** — not just neighbors.
- Uses **Dijkstra's algorithm** to compute the shortest-path tree with itself as root $\rightarrow$ derives its own routing table (each router's tree/table is unique even though the underlying topology data — the Link-State Database, LSDB — is identical everywhere).
- No count-to-infinity problem (each router computes routes independently from full, consistent topology data) — converges faster, but at higher bandwidth/CPU/memory cost.
- **TRAP:** LSR floods to the *entire network* about a router's *own direct neighbors only*; DVR tells *neighbors only* about the *entire network* (as the router understands it) — these are inverse in scope.

9.5 OSPF (Open Shortest Path First) — a real LSR implementation

- **Intra-domain** protocol; metric = flexible **cost** (can reflect bandwidth/throughput/RTT/reliability — administrator-configurable), unlike RIP's fixed hop count.
- Divides a large AS into **Areas** connected via a central **Backbone Area**, to limit flooding scope.
- Runs **directly over IP** (not TCP/UDP).
- Faster convergence and more robust than RIP (a single router's failure has less network-wide impact), at the cost of more complex/heavier update traffic.

9.6 DVR vs LSR vs RIP vs OSPF

Feature	DVR (RIP)	LSR (OSPF)
Shares	Own routing table (dest+cost)	Own direct-link states (LSPs)
Shared with	Neighbors only	All routers (flooded)
Knowledge basis	Local	Global (full topology)
Algorithm	Bellman-Ford	Dijkstra
Convergence	Slow	Fast
Bandwidth/traffic use	Low	High
Count-to-infinity?	Yes (mitigated by split horizon)	No
Era popularized	~1980s	~1990s
Metric	Hop count (max 15)	Configurable cost

9.7 Routing in Subnets (Longest Prefix Match)

- Router does **bitwise AND** of destination IP with each routing-table subnet mask, compares to table entries:
- **Single match** $\rightarrow$ forward via that interface.
- **Multiple matches** $\rightarrow$ use the entry with the **longest subnet mask** (longest prefix match).
- **No match** $\rightarrow$ use the **default route**.

- **Fixed-Length Subnetting:** one mask for all subnets, AND done once. **Variable-Length Subnetting:** different masks per subnet, AND done against each until a match is found.

Questions

Q1. [Trap] In Distance Vector Routing, a router shares its routing information with:

A. All routers in the network B. Only its immediate neighbors C. Only the backbone router D. No one; it is passive

Answer: B. Why: DVR is based on local knowledge — each router exchanges its distance vector only with directly connected neighbors, unlike Link State Routing which floods to everyone.

Q2. The "count-to-infinity" problem arises in which routing approach, and is reduced by which technique?

A. Link State Routing; reduced by Dijkstra's algorithm B. Distance Vector Routing; reduced by Split Horizon C. OSPF; reduced by flooding D. Flooding; reduced by hop count

Answer: B. Why: Count-to-infinity is a DVR-specific instability from stale neighbor information; Split Horizon (not re-advertising a route back to the neighbor it came from) mitigates it.

Q3. In RIP, a destination reported with hop count 16 is interpreted as:

A. 16 hops away, still reachable B. Unreachable (infinity) C. The default route D. A routing loop

Answer: B. Why: RIP caps usable hop counts at 15; a metric of 16 is defined as "unreachable," limiting RIP to small/medium networks.

Q4. Which algorithm underlies Link State Routing's shortest-path computation, and which underlies Distance Vector Routing?

A. Dijkstra for DVR, Bellman-Ford for LSR B. Bellman-Ford for both C. Dijkstra for LSR, Bellman-Ford for DVR D. Neither uses a named algorithm

Answer: C. Why: LSR builds a shortest-path tree via Dijkstra's algorithm using full topology knowledge; DVR is rooted in the Bellman-Ford relaxation approach using only neighbor-exchanged distances.

Q5. Why does OSPF generally converge faster and prove more robust than RIP for large networks?

A. OSPF uses hop count exclusively B. OSPF floods full topology (link-state) data so each router independently computes consistent shortest paths, avoiding DVR-style instabilities C. OSPF has a lower maximum hop count D. OSPF does not use any routing algorithm

Answer: B. Why: Because every OSPF router works from a complete, consistent view of the topology (via LSPs + Dijkstra), it avoids the slow, locally-inconsistent convergence that plagues distance-vector protocols like RIP.

Q6. A router's routing table has two matching entries for a destination IP after the bitwise-AND check, with subnet masks /24 and /28. Which entry is used?

A. /24, since it was likely added first B. /28, the longest prefix match C. Both are used via load balancing D. Neither; the default route is used

Answer: B. Why: When multiple entries match, the router selects the one with the longest subnet mask (most specific match) — the standard longest-prefix-match rule.

Q7. OSPF's metric differs from RIP's in that OSPF:

A. Always equals hop count like RIP B. Can be configured based on factors like bandwidth/throughput/reliability, not just a fixed per-hop cost C. Is limited to a maximum of 15 D. Requires UDP port 520 like RIP

Answer: B. Why: RIP assigns a uniform "1" per hop; OSPF allows administrator-defined cost reflecting actual link characteristics, giving more realistic path selection.

Q8. [Trap] "Link State Routing tells each router what it knows about the entire network." Correct this statement:

A. It is correct as stated B. Link State Routing has each router tell the WHOLE network what it knows about its OWN direct links only C. LSR never shares any information D. LSR only works within a single subnet

Answer: B. Why: The scope is inverted from DVR: in LSR, a router broadcasts (floods) information about only its own immediate links, but this reaches every router in the network — the opposite pattern from DVR (local broadcast of global-ish knowledge).

10. Transport Layer — Services & TCP

10.1 Transport Layer Role & Protocol Types

- Network layer = **host-to-host** delivery. Transport layer = **process-to-process** delivery (the missing link to the actual application).
- **Connectionless (UDP)**: each segment independent, no setup/teardown, unreliable, fast.
- **Connection-oriented (TCP, SCTP)**: connection established before transfer, torn down after; reliable (flow + error control done **end-to-end**, not link-by-link like DLL).
- **TCP**: connection-oriented, reliable, full-duplex, byte-stream. **SCTP**: connection-oriented, reliable, combines TCP+UDP features (message-oriented + multi-streaming).

10.2 Ports & Socket Addresses

- Port number = **16-bit** (0–65535), identifies a process on a host.

Well-Known Ports:	0 – 1023	(IANA-assigned; e.g. HTTP 80, HTTPS 443, FTP 20/21, SSH 22, Telnet 23, SMTP 25, DNS 53, DHCP 67/68, POP3 110)
Registered Ports:	1024 – 49151	(registerable, not IANA-controlled)
Dynamic/Ephemeral:	49152 – 65535	(temporary, client-side, randomly chosen)

- **Socket address** = **IP address** + **Port number**. A TCP connection is uniquely identified by the pair of socket addresses (client socket, server socket).
- **Encapsulation**: App message + (src,dst) socket addresses → Transport layer adds header → **segment**. **Decapsulation** reverses this at the receiver.

10.3 TCP Header (20–60 bytes)

Src Port(16b) Dst Port(16b)
Sequence Number (32b)
Acknowledgment Number (32b)
HdrLen(4b) Reserved(6b) Flags(6b: URG,ACK,PSH,RST,SYN,FIN) Window Size(16b)
Checksum(16b) Urgent Pointer(16b)
Options (0-40 bytes)

- **Header Length field (4 bits)** = number of 4-byte words → value ranges **5 to 15** → header size = value × 4 bytes → **20 bytes (no options) to 60 bytes (max options, 40B)**.
- **Checksum** covers the whole TCP segment + a **pseudo-header from IP** (pseudo-header's Protocol field = **6** for TCP).
- **Window field** = receiver's advertised buffer space, max **65,535 bytes** (16 bits) — can be extended via a **Window Scale option** (up to ~1 GB, effectively 30 bits) since 64KB is tiny for modern high-bandwidth links.
- **Flags**: **PSH** = push data to application immediately, skip buffering (used for interactive/chat traffic). **RST** = abort an invalid/erroneous connection. **URG** + Urgent Pointer = marks urgent data.
- **TRAP**: TCP counts **bytes**, not packets (unlike IP, which counts packets) or bits (unlike DLL/HDLC, which counts bits).

10.4 Sequence & Acknowledgment Numbers

- **ISN (Initial Sequence Number)** is chosen **randomly** at connection setup (not 0) — prevents old-duplicate-segment confusion and improves security.
- Next segment's sequence number = previous segment's sequence number + bytes it carried.
- **Acknowledgment number** is **cumulative**: ACK = (last byte received safely) + 1.

Worked Example

5000-byte file sent as five 1000-byte segments; first byte numbered **10001**.
Segment sequence numbers: **10001, 11001, 12001, 13001, 14001**.

Wrap-Around

Wrap-Around Time (WAT) = $2^{32} / \text{Bandwidth (bytes/sec)}$

- If **WAT > segment Lifetime (LT, ~180s / 3 min)** → safe (old sequence numbers fully retired before reuse). If **WAT < LT** → risk of duplicate/confusing sequence numbers on very high-bandwidth links.
- **TRAP**: Higher bandwidth → **smaller** WAT (sequence space exhausts faster) → *more* risk of wrap-around conflicts, not less.

10.5 Connection Establishment & Termination

- **3-Way Handshake (establishment)**: Client --SYN--> Server --SYN+ACK--> Client --ACK--> Server. SYN (and FIN) segments **consume one sequence number** each even with no data payload.
- **Termination**: initiator sends **FIN**; other side replies **FIN+ACK** (or separately ACK then its own independent FIN) → final **ACK**. **TRAP**: Textbooks often show termination as a 4-way exchange (each side sends its own independent FIN, since TCP is full-duplex and each direction must close separately) — the "3-segment" version shown here just combines the responder's ACK and FIN into one segment; do not assume termination is always exactly symmetric with the 3-way SYN handshake.
- **SYN Flooding (DoS)**: attacker sends many spoofed-source SYNs; server allocates resources (TCB) for each and waits for the final ACK that never comes → resource exhaustion, denial of service to real clients.

10.6 Retransmission Triggers

Trigger	Confidence of loss
Timeout timer expiry	Strong signal of congestion — nothing came back at all
3 duplicate ACKs ("Fast Retransmit")	Weaker signal — later segments <i>are</i> getting through, only one was likely dropped

• TCP effectively blends **Go-Back-N** (cumulative ACKs, simplicity) with **Selective Repeat** (accepts/buffers out-of-order segments) behavior.

Questions

Q1. [Numerical] A TCP segment's header length field value is 10. What is the actual header size?

A. 10 bytes B. 20 bytes C. 40 bytes D. 60 bytes

Answer: C. **Why:** Header length = field value $\times$ 4 bytes = $10 \times 4 = 40$ bytes.

Q2. Why does TCP choose a random Initial Sequence Number (ISN) instead of starting at 0?

A. To save memory B. To prevent confusion with old duplicate segments from a previous connection and improve security C. Because 0 is reserved D. To simplify checksum calculation

Answer: B. **Why:** A predictable/reused starting point could let old delayed segments be mistaken for new ones, or allow sequence-prediction attacks.

Q3. [Trap] Which flag should be set for a chat application needing immediate delivery without buffering delay?

A. RST B. URG C. PSH D. FIN

Answer: C. **Why:** PSH forces the transport layer to send immediately and tells the receiver to hand data to the application without waiting to fill a full segment.

Q4. [Numerical] Network bandwidth = 2^{32} bytes/sec (hypothetical). What is the sequence number wrap-around time?

A. 1 second B. 2 seconds C. 4 seconds D. It never wraps around

Answer: A. **Why:** $WAT = 2^{32} / \text{Bandwidth} = 2^{32} / 2^{32} = 1$ second.

Q5. [Trap] A sender detects segment loss via 3 duplicate ACKs rather than a timeout. Compared to timeout-triggered loss detection, this indicates:

A. A stronger, more certain sign of network congestion B. A weaker sign of congestion, since later segments are still getting through C. That the connection has failed completely D. No difference at all

Answer: B. **Why:** Duplicate ACKs mean subsequent segments ARE arriving (just one was dropped/reordered) — a much milder signal than a full timeout where nothing arrived at all.

Q6. In the TCP pseudo-header used for checksum calculation, the Protocol field value for TCP is:

A. 17 B. 6 C. 1 D. 89

Answer: B. **Why:** Protocol number 6 identifies TCP in the IP pseudo-header (17 = UDP, 1 = ICMP, 89 = OSPF).

Q7. What is the maximum receiver window size TCP can advertise using only the standard 16-bit Window field (without the Window Scale option)?

A. 4095 bytes B. 65,535 bytes C. 1 GB D. Unlimited

Answer: B. **Why:** 16 bits $\rightarrow$ max value 65,535 bytes; larger windows need the Window Scale option (extends effectively to ~ 30 bits / $\sim 1\text{GB}$).

Q8. Which TCP behavior most resembles Go-Back-N, and which resembles Selective Repeat?

A. Cumulative ACKs resemble GBN; buffering/accepting out-of-order segments resembles SR B. TCP uses only Stop-and-Wait C. TCP uses only Selective Repeat, never GBN D. TCP uses only Go-Back-N, never SR

Answer: A. **Why:** TCP's cumulative acknowledgment scheme is GBN-like, while its ability to buffer and accept out-of-order segments (rather than discarding them) is SR-like — a hybrid design.

11. Transport Layer — Congestion Control & UDP

11.1 Sender Window = min(Receiver Window, Congestion Window)

Sender Window Size = MIN(Receiver Window Size, Congestion Window Size)

- **Receiver window (flow control):** advertised by receiver via TCP header; caps data in flight so the receiver isn't overwhelmed.
- **Congestion window (cwnd):** sender-side-only estimate (never transmitted on the wire) of what the *network* can currently handle without congesting.
- TCP has 4 windows total for bidirectional communication (send+receive window, ×2 directions).

11.2 TCP Congestion Control — 3 Phases

Phase 1 - Slow Start (EXPONENTIAL growth):

cwnd starts at 1 MSS

After each RTT: $cwnd = cwnd \times 2$ (1, 2, 4, 8, 16 MSS ... after RTT 1,2,3,4...)

Continues until cwnd reaches ssthresh (slow-start threshold)

$ssthresh = (\text{Receiver Window Size} / \text{MSS}) / 2$

Phase 2 - Congestion Avoidance (LINEAR / additive growth):

On each ACK: $cwnd = cwnd + 1 \text{ MSS}$ (roughly +1 MSS per RTT)

Continues until cwnd = Receiver Window Size, or loss detected

Phase 3 - Congestion Detection (on loss):

If detected via TIMEOUT (strong congestion signal):

$ssthresh = cwnd / 2$; $cwnd = 1 \text{ MSS}$; restart Slow Start

If detected via 3 DUPLICATE ACKs (weak signal, "Fast Retransmit/Recovery"):

$ssthresh = cwnd / 2$; $cwnd = ssthresh$; resume Congestion Avoidance (NOT slow start)

- **TRAP:** "Slow Start" is a **misnomer** — its growth is exponential (fastest phase), not slow; it's only "slow" relative to just blasting the full window immediately.
- **TRAP:** Timeout and Triple-Duplicate-ACK loss detection do **NOT** react the same way — timeout drops all the way back to cwnd=1 MSS (full Slow Start restart); 3-dup-ACK only drops to ssthresh and resumes linear growth (Fast Recovery) — because 3-dup-ACK is weaker evidence of congestion (later packets are still arriving).
- Overall behavior = **AIMD** (Additive Increase, Multiplicative Decrease).

11.3 TCP Timers

Timer	Purpose	Behavior
Time-Wait	Avoid conflicts with late/duplicate packets after connection close	Waits 2 × segment Lifetime (2×LT) before fully releasing the port
Keep-Alive	Detect & close idle/dead connections	Sends up to 10 probes at 75-second intervals ; closes if no reply
Persistent	Avoid deadlock when receiver advertises window size = 0	Sender periodically sends a 1-byte probe to check if window has reopened
Acknowledgment (delayed ACK)	Efficient cumulative ACKs via piggybacking	Starts on segment arrival; ACKs everything received within the window in one shot

11.4 RTT / Timeout-Timer (TOT) Estimation Algorithms

- Setting TOT is harder at Transport layer than DLL (DLL only deals with one adjacent hop; Transport spans multi-hop, variable paths).
- **Rule of thumb:** High traffic → RTT rising → increase TOT (else false retransmissions worsen congestion). Low traffic → RTT falling → decrease TOT (else sender waits too long).

Basic Algorithm: $IRTT(n+1) = \alpha \times IRTT(n) + (1-\alpha) \times ARTT(n)$; $TOT = 2 \times IRTT$

Jacobson's Algorithm adds deviation:

$ID(n+1) = \alpha \times ID(n) + (1-\alpha) \times AD(n)$ where $AD(n) = |IRTT(n) - ARTT(n)|$

$TOT = 4 \times ID + IRTT$ (accounts for RTT *variability*, not just a fixed ×2 multiplier)

Karn's Modification: ignore RTT samples from RETRANSMITTED segments (their true RTT is ambiguous); instead simply DOUBLE the timeout timer on each retransmission.

- **TRAP:** The Basic Algorithm's flat "×2" multiplier has no adaptive justification — Jacobson's fixes this by explicitly modeling RTT deviation, giving a more responsive, accurate timeout.

11.5 Silly Window Syndrome (SWS)

- Problem: sender sends tiny segments repeatedly, or receiver advertises tiny window updates repeatedly → terrible efficiency (huge header-to-data overhead ratio).
- **Nagle's Algorithm** (sender-side fix): send the first small segment immediately, then **buffer** further data until the first is ACKed; then send buffered data as one segment. (Bad for latency-sensitive apps — can be disabled, e.g. TCP_NODELAY.)
- **Clark's Solution** (receiver-side fix): don't advertise a window update for a tiny amount of newly-free space — wait until a **substantial** amount (e.g., half the buffer) is free.
- **TRAP:** Nagle's Algorithm and Clark's Solution attack **opposite ends** of the same problem (sender-side batching vs receiver-side window-update batching) — they are complementary, not alternatives.

11.6 UDP (User Datagram Protocol)

```

UDP Header (fixed 8 bytes):
| Source Port (16b) | Destination Port (16b) |
| Total Length (16b) | Checksum (16b) |

```

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Reliable (ACK, retransmission)	Unreliable (best-effort)
Ordering	Ordered (sequence numbers)	No ordering guarantee
Header size	20–60 bytes	Fixed 8 bytes
Flow/Congestion control	Yes	None
Overhead	Higher	Very low
Typical use	File transfer, web (HTTP), email	DNS, streaming/VoIP, DHCP, RIP, SNMP

- UDP is preferred where **speed/low latency matters more than perfect delivery** (real-time media tolerates minor loss better than delay), or where the application implements its own reliability (e.g., FTP).

Questions

Q1. [Numerical] cwnd starts at 1 MSS. What is cwnd after 4 RTTs of Slow Start (no loss)?

A. 4 MSS B. 8 MSS C. 16 MSS D. 32 MSS

Answer: C. **Why:** Exponential growth: $cwnd = 2^4 \text{RTT} = 2^4 = 16 \text{ MSS}$.

Q2. [Trap] TCP detects a segment loss via a timeout (not 3 duplicate ACKs). What happens to cwnd and ssthresh?

A. ssthresh = cwnd/2, cwnd = ssthresh, resume Congestion Avoidance B. ssthresh = cwnd/2, cwnd = 1 MSS, restart Slow Start C. Both cwnd and ssthresh double D. No change, TCP ignores timeouts

Answer: B. **Why:** Timeout is a strong congestion signal → full reset: cwnd drops to 1 MSS and Slow Start restarts (unlike the milder 3-dup-ACK case, which only halves and resumes linear growth).

Q3. Which timer prevents a "zero window" deadlock where the receiver's window has silently reopened but the sender never finds out?

A. Keep-Alive Timer B. Time-Wait Timer C. Persistent Timer D. Acknowledgment Timer

Answer: C. **Why:** The Persistent Timer triggers periodic 1-byte probes specifically to detect a reopened window after a zero-window advertisement.

Q4. [Trap] What is the actual growth pattern during TCP's "Slow Start" phase?

A. Linear (+1 MSS per RTT) B. Exponential (doubles per RTT) C. Logarithmic D. Constant (no growth)

Answer: B. **Why:** Despite the name, Slow Start grows exponentially (cwnd doubles each RTT) until ssthresh is reached — a classic naming trap.

Q5. Why does Karn's Modification ignore RTT measurements taken from retransmitted segments?

A. Retransmitted segments are always corrupted B. It's ambiguous whether the received ACK corresponds to the original or the retransmitted copy, making the measured RTT unreliable C. Retransmissions never receive ACKs D. To save computation

Answer: B. **Why:** If a segment is retransmitted, an ACK could be for either copy — using this ambiguous timing to update RTT estimates would corrupt future timeout calculations.

Q6. [Numerical] What is the fixed size of a UDP header?

A. 4 bytes B. 8 bytes C. 20 bytes D. 16 bytes

Answer: B. **Why:** UDP header = Source Port + Dest Port + Length + Checksum, each 16 bits = 8 bytes total (much smaller than TCP's 20–60 byte header).

Q7. Which mechanism specifically prevents a sender from repeatedly transmitting tiny, inefficient segments?

A. Clark's Solution B. Nagle's Algorithm C. Keep-Alive Timer D. Karn's Modification

Answer: B. **Why:** Nagle's Algorithm buffers outgoing small segments until the previous one is acknowledged, then sends accumulated data as one larger segment.

Q8. Why is UDP preferred over TCP for DNS queries and live video/audio streaming?

A. UDP guarantees delivery, TCP does not B. UDP's lower overhead and lack of retransmission delay suit latency-sensitive, loss-tolerant traffic better than TCP's reliability guarantees C. UDP has a larger header for more metadata D. TCP cannot be used for these applications at all

Answer: B. **Why:** Real-time applications tolerate occasional packet loss far better than the added latency TCP's acknowledgment/retransmission machinery would introduce.

Q9. [Numerical] Receiver window size = 8000 bytes, MSS = 1000 bytes. What is the Slow Start threshold (ssthresh)?

A. 2 B. 4 C. 8 D. 16

Answer: B. **Why:** $ssthresh = (\text{Receiver Window} / \text{MSS}) / 2 = (8000/1000)/2 = 8/2 = 4$.

12. Application Layer Protocols

12.0 Consolidated Port/Protocol Reference (memorize this table)

Protocol	Port(s)	Transport	Purpose
SMTP	25	TCP	Sending/pushing mail (server→server, client→server)
POP3	110	TCP	Retrieving mail — downloads then typically deletes from server
IMAP4	143	TCP	Retrieving mail — keeps mail on server, multi-device sync, folders
HTTP	80	TCP	Web page transfer (stateless)
HTTPS	443	TCP (+SSL/TLS)	Encrypted web transfer
FTP	21 (control) + 20 (data)	TCP	File transfer — TWO separate connections
Telnet	23	TCP	Remote login, plaintext (insecure)
SSH	22	TCP	Secure remote login — replaced Telnet
DNS	53	UDP (queries) / TCP (zone transfer)	Domain name ↔ IP resolution
DHCP	67 (server) / 68 (client)	UDP	Dynamic IP address assignment

- **TRAP:** FTP is the classic "two ports" exam question — control (21) stays open for commands, data (20) opens per-transfer. Don't answer "FTP uses port 21" alone.
- **TRAP:** SMTP only **sends/pushes** mail; POP3/IMAP4 are **pull** protocols used only for retrieval — SMTP is never used to download mail.

12.1 Email — SMTP, POP3, IMAP4

- Email is **store-and-forward**: sender/receiver need not be online simultaneously.
- **MTA (Message Transfer Agent)**: handles transfer using SMTP, used **twice** — sender→sender's mail server, then sender's server→receiver's server.
- **POP3**: simple; **Delete mode** (mail removed from server after download — for a permanent PC) vs **Keep mode** (mail stays on server — for accessing from multiple/temporary devices). Cannot organize mail in folders on the server or partially download.
- **IMAP4**: more powerful — supports server-side folders/hierarchies, partial download, header preview before download, search before download.
- **MIME**: extends email beyond plain 7-bit ASCII to support non-English scripts, binary/audio/video attachments (converts non-ASCII → ASCII for transport, then back).
- **Webmail (Hotmail/Gmail-style)**: browser→mail server uses **HTTP** (not POP/IMAP) for both sending and retrieval; server→server transfer between mail servers still uses **SMTP**.
- Email address rule: exactly one "@"; no spaces; case-insensitive.
- **Bcc rule**: Bcc recipients are invisible to To/Cc and to each other; a Bcc'd person CAN see the To/Cc list.

12.2 WWW & Cookies

- WWW (Tim Berners-Lee, 1989, CERN) = client-server system of linked web pages.
- **URL** has 4 parts: **Protocol : Host : Port : Path** (e.g., http://host:port/path).
- WWW/HTTP is fundamentally **stateless** — each request/response is independent, server doesn't remember prior visits.
- **Cookies** solve statelessness: server sends a cookie in its response → browser stores it (keyed by domain) → browser automatically re-sends it on future requests to that domain → server recognizes the returning client.
- **TRAP:** A cookie's contents are set and read only by the **server** — the browser stores but never interprets/discloses cookie contents to the user ("made by the server, eaten by the server").

12.3 HTTP / HTTPS

- HTTP = connection-oriented over **TCP port 80**; defines request/response rules between browser and server.
- **Non-persistent connection**: new TCP connection per request/response (overhead for pages with many embedded objects).
- **Persistent connection (default since HTTP/1.1)**: one TCP connection reused for multiple requests — saves setup time/RTTs.
- **Proxy server**: caches responses to reduce load/latency for repeated requests.
- **HTTPS** = HTTP + SSL/TLS, **port 443** — adds Confidentiality (encryption), Integrity, Authentication (via digital certificates). HTTP itself has none of these.
- **TRAP:** HTTP uses a **single** TCP connection for a request/response cycle — unlike FTP's separate control+data connections.

12.4 FTP, DNS, Telnet, DHCP

- **FTP**: two TCP connections — **Control (port 21)** for commands, **Data (port 20)** for the actual file transfer. Sends credentials/data in **plaintext** (insecure) unless upgraded to SSL-FTP.
- **DNS**: translates domain names → IP addresses; **hierarchical & distributed**: Root servers → TLD servers (.com, .in, .org) → Authoritative servers (final name→IP mapping). Domain name space = inverted tree, max 128 levels, each label ≤ 63 characters, root = null label.
- **Generic domains**: .com, .edu, .mil, .org, .net. **Country domains**: .in, .uk, .us. **Inverse domain**: IP → name (reverse lookup).
- **Telnet**: port 23, remote virtual-terminal login; **plaintext** (major security flaw) → replaced by **SSH (port 22)** for production use; still used for raw port/connectivity testing.
- **DHCP**: Application-layer protocol using **UDP**; automates IP assignment via the **DORA** process: **Discover** (client broadcasts) → **Offer** (server proposes an IP) → **Request** (client accepts) → **Acknowledge** (server confirms). Server = port 67, Client = port 68.
- **0.0.0.0** = "I have no IP yet" (placeholder during Discover). **127.0.0.1** = loopback/localhost (tests own NIC, unrelated to DHCP). **169.254.x.x (APIPA)** = self-assigned address when DHCP server is unreachable.

12.5 VoIP & Mobile Communication Protocols

- **VoIP**: converts voice → digital IP packets (packet-switched) instead of the old PSTN's dedicated circuit per call.
- **SIP (Session Initiation Protocol)**: the "manager" — sets up/rings/tears down the call (signaling).

- **RTP (Real-time Transport Protocol):** the "courier" — carries the actual voice/video media stream.
- **PPP (Point-to-Point Protocol):** Data Link layer; direct link between 2 nodes (dial-up/DSL/PPPoE); supports authentication via **PAP** and **CHAP**; carries multiple network-layer protocols (multiprotocol encapsulation) — unlike Ethernet, which needs no login.
- **Mobile generations:** 1G (analog voice) → **2G/GSM** (digital voice+SMS, uses **TDMA**, introduced the **SIM card**) → **2.5G/GPRS** (packet-switched data bridge to 3G, "always-on," ~114 kbps) → 3G (mobile internet, video calls) → 4G/LTE (all-IP, HD streaming) → 5G (ultra-low latency, massive IoT).
- **WLL (Wireless Local Loop):** replaces the physical "last-mile" copper wire with a fixed wireless radio link to the local exchange — used in rural/hard-to-cable areas.
- **TRAP:** GSM uses **circuit switching** for voice (via TDMA time-slots); GPRS was the shift to **packet switching** for data — don't assume all mobile-gen protocols are packet-switched from the start.

12.6 Web Services (Search, HTML/XML, Hosting)

- **3 pillars of the Web:** URL (addressing) + HTTP (transfer protocol) + HTML (page structure/display).
| Feature | HTML | XML |
| ---|---|---|
| Purpose | Document **display/formatting** | Data **transport/storage** (structure & meaning) |
| Tags | Fixed, predefined (e.g. `<h1>`, `<img>`) | Custom, user-defined (e.g. `<price>`) |
| Concerned with | How content looks | What the data means |
- **Static website:** same content for every visitor (pure HTML/CSS). **Dynamic website:** content generated per-user via server-side scripting + database (e.g., PHP/Python).
- **Web server:** software (Apache, Nginx) that stores site files and serves them on request. **Web hosting:** the rented service/infrastructure that keeps a web server online.
- Search engine (Google, Bing) ≠ Web browser (Chrome, Safari): browser is the client app; search engine is a website you visit through the browser.

Questions

Q1. [Trap] FTP is best described as using:

- A. A single TCP connection on port 21 for everything B. Two separate connections — control (port 21) and data (port 20) C. UDP on port 21 D. A single UDP connection on port 20

Answer: B. **Why:** FTP uniquely separates command/control traffic (port 21) from the actual file data transfer (port 20).

Q2. Which protocol is used only to send/push email, never to retrieve it?

- A. POP3 B. IMAP4 C. SMTP D. HTTP

Answer: C. **Why:** SMTP is a push protocol for sending mail between clients and servers; retrieval requires a pull protocol like POP3 or IMAP4.

Q3. A user wants to read email from multiple devices with all messages and folders staying synced on the server. Which protocol should they use?

- A. POP3 in delete mode B. POP3 in keep mode C. IMAP4 D. SMTP

Answer: C. **Why:** IMAP4 keeps mail organized in server-side folders and supports multi-device sync; POP3 (especially delete mode) is designed for single-device, download-then-remove access.

Q4. [Trap] The World Wide Web (HTTP) is inherently stateless. Which mechanism allows a server to recognize a returning client across requests?

- A. TCP sequence numbers B. Cookies C. The URL path D. DNS caching

Answer: B. **Why:** Cookies are stored client-side and automatically resent to the originating domain, letting an otherwise stateless server infer session continuity.

Q5. HTTPS differs from HTTP mainly by adding which capability, via SSL/TLS on port 443?

- A. Faster page loading only B. Confidentiality, Integrity, and Authentication C. A second TCP connection like FTP D. Persistent connections (HTTP lacks these)

Answer: B. **Why:** HTTPS wraps HTTP traffic in an encrypted SSL/TLS tunnel, adding the three security properties HTTP itself lacks; both HTTP/1.1 and HTTPS support persistent connections.

Q6. In DNS's hierarchical structure, which server holds the FINAL mapping from a specific domain name to its IP address?

- A. Root server B. TLD server C. Authoritative name server D. Local resolver cache only

Answer: C. **Why:** Root servers redirect to TLD servers (.com, .in, etc.), which in turn point to the Authoritative server that actually stores the definitive name→IP record.

Q7. Telnet has been largely replaced in production use by which more secure protocol?

- A. FTP B. SSH C. HTTP D. SNMP

Answer: B. **Why:** Telnet transmits credentials/data in plaintext; SSH (port 22) provides the same remote-terminal function with encryption.

Q8. [Numerical/Conceptual] In the DHCP DORA process, what is the correct order of messages?

- A. Offer → Discover → Request → Acknowledge B. Discover → Offer → Request → Acknowledge C. Discover → Request → Offer → Acknowledge D. Request → Discover → Acknowledge → Offer

Answer: B. **Why:** Client broadcasts Discover → server replies with an Offer → client sends Request to accept it → server sends Acknowledge to finalize.

Q9. Which IP address indicates "DHCP server unreachable, self-assigned fallback address"?

- A. 0.0.0.0 B. 127.0.0.1 C. 169.254.x.x (APIPA) D. 255.255.255.255

Answer: C. **Why:** APIPA (169.254.x.x) is auto-assigned by Windows when no DHCP server responds; 0.0.0.0 means "no IP yet," and 127.0.0.1 is the loopback address (unrelated to DHCP failure).

Q10. GSM's core multiple-access technology, allowing several users to share one frequency channel by taking turns in time, is:

- A. CDMA B. FDMA C. TDMA D. OFDM

Answer: C. **Why:** GSM (2G) uses TDMA, slicing time into slots so up to 8 users share one radio channel.

Q11. [Trap] GPRS's key innovation over plain GSM was:

- A. Introducing the SIM card B. Switching from circuit-switched voice to packet-switched, "always-on" data C. Introducing video calling D. Introducing digital voice for the first time

Answer: B. **Why:** GSM handled only circuit-switched voice; GPRS ("2.5G") added packet-switched, always-on data — the bridge to 3G.

Q12. The key structural difference between HTML and XML is that:

- A. HTML is for data storage, XML is for display B. HTML uses fixed display tags; XML uses custom, user-defined tags to describe data meaning C. XML cannot be used on the web D. They are functionally identical

Answer: B. **Why:** HTML tags (`<h1>`, `<img>`) control visual presentation; XML tags are custom-defined to describe the structure/meaning of data for exchange between systems.

Q13. Which VoIP protocol is responsible for setting up, ringing, and tearing down a call, as opposed to carrying the actual voice data?

- A. RTP B. SIP C. TCP D. HTTP

Answer: B. **Why:** SIP handles call signaling (setup/teardown); RTP carries the real-time media stream itself.

Q14. PPP's key advantage over a plain Ethernet connection for a dial-up/DSL link is:

- A. Higher raw bandwidth B. Built-in authentication (PAP/CHAP) before granting network access C. Use of MAC addresses D. Native encryption of all data

Answer: B. **Why:** PPP requires the far end to authenticate (PAP or the stronger CHAP) before a connection is established — Ethernet has no such built-in login step.

13. Networking Hardware Devices

13.1 Layer-by-Layer Device Summary

Device	OSI Layer	Uses which address?	Filters/forwards intelligently?	Collision domain	Broadcast domain
Repeater	Physical (L1)	None	No — regenerates signal only	Not separated	Not separated
Hub	Physical (L1)	None (multiport repeater)	No — broadcasts to all ports	One single domain for all ports	Not separated
Bridge	Data Link (L2)	MAC address	Yes — has a MAC address table	Separates collision domains per segment	NOT separated (broadcasts pass through)
Switch	Data Link (L2)	MAC address	Yes — multiport bridge, per-port MAC table	Each port = own collision domain	NOT separated (unless VLANs used)
Router	Network (L3) (also L1/L2)	IP address	Yes — routing table, no collisions	Separates	Separates broadcast domains
Gateway	Any layer (L1–L7)	Protocol-dependent	Yes — protocol conversion	—	—

13.2 Key Facts

- **Repeater:** regenerates (not merely amplifies) a weakened signal to extend cable distance; blind to frames/addresses.
- **Hub = multiport repeater:** all ports share ONE collision domain (half-duplex); no MAC table, no filtering — data sent to one port is repeated to ALL ports.
- **Bridge:** connects LAN segments of the *same* technology; builds a MAC address table to filter frames onto only the correct segment — reduces collisions, but still forwards broadcasts everywhere.
- **Switch:** essentially a fast, multiport bridge — every port is its own collision domain, but the whole switched LAN is still one broadcast domain (until VLANs subdivide it).
- **Router:** operates using IP addresses; reads the routing table to forward packets between different networks; does routing/filtering/forwarding — **routers separate broadcast domains** (this is the key differentiator from switches).
- **Gateway:** connects networks using **different protocols** (protocol conversion); can operate at any OSI layer; broader role than router — e.g., protocol translation, proxy, NAT, firewall, deep packet inspection.
- **TRAP:** "Bridge/Switch separate collision domains but NOT broadcast domains; only Routers separate broadcast domains" — this is one of the most frequently tested hardware facts.
- **TRAP:** A **Hub** does NOT separate even collision domains (all ports = 1 domain) — weaker than a Bridge/Switch, which separates collision domains per port/segment.

Questions

Q1. Which device operates purely at the Physical layer and has no concept of MAC addresses?

A. Bridge B. Switch C. Hub D. Router

Answer: C. **Why:** A Hub (multiport repeater) only regenerates and repeats electrical signals; it has no MAC address table and floods all ports.

Q2. [Trap] Which of these devices separates collision domains but NOT broadcast domains?

A. Router B. Hub C. Switch D. Repeater

Answer: C. **Why:** A Switch gives each port its own collision domain but still forwards broadcast frames to every port — only a Router separates broadcast domains.

Q3. A device that connects two networks running entirely different protocols (e.g., converting between two different network architectures) is best described as a:

A. Repeater B. Hub C. Gateway D. Bridge

Answer: C. **Why:** Gateways specifically handle protocol conversion between dissimilar networks and can function at any OSI layer, unlike routers/bridges which typically operate within a single protocol family.

Q4. Why is a switch generally preferred over a hub in modern LANs?

A. A switch is cheaper B. A switch gives every port its own collision domain via MAC-address-based forwarding, reducing collisions C. A switch works only at the Physical layer D. A switch cannot operate in full-duplex

Answer: B. **Why:** Unlike a hub's shared collision domain, a switch's per-port forwarding (via its MAC table) means devices on different ports don't collide with each other.

Q5. Which device makes its forwarding decisions using the destination IP address and a routing table?

A. Switch B. Bridge C. Router D. Repeater

Answer: C. **Why:** Routers operate at the Network layer, using IP addresses and routing tables/policies to select the next hop — unlike Bridges/Switches, which use MAC addresses at Layer 2.

14. Network Security Fundamentals & Cryptography

14.1 Security Goals (CIA Triad +2) — OSI Security Architecture (ITU-T X.800)

- **Confidentiality** — only authorized users read data. Tools: Encryption, Access Control, VPN, SSL/TLS.
- **Integrity** — data unaltered/detectable if changed. Tools: Hash, MAC, Digital Signature.
- **Availability** — resources accessible when needed. Threats: DoS/DDoS, hardware/power failure. Tools: redundancy, load balancing, disaster recovery.
- **Authentication** ("who are you?") — Peer-entity (session start) vs Data-origin (per-message sender verification).
- **Non-Repudiation** — sender/receiver can't deny an action. **NRO** = proof of origin, **NRR** = proof of receipt.
- **TRAP**: Cryptography alone gives Confidentiality, Integrity, Authentication, Non-Repudiation — **NOT Availability** (needs redundancy/DoS defenses, not crypto).

14.2 Security Attacks

Feature	Passive Attack	Active Attack
Data modified?	No	Yes
Detection	Difficult	Easier
Prevention	Easier (encryption)	Harder
Target	Confidentiality	Integrity & Availability
Types	Release of message contents, Traffic Analysis	Masquerade, Replay, Modification, DoS

- **Traffic Analysis**: even encrypted traffic leaks metadata (who/when/how much) — encryption does NOT stop this.
- **Replay**: resend a captured *valid* message later. **Masquerade**: impersonate using stolen credentials. **Modification**: alter message content in transit.

14.3 Security Services (X.800)

Authentication, Access Control ("what are you allowed to do," comes after authentication), Data Confidentiality, Data Integrity, Non-Repudiation.

Guarantee ladder: Hash Function -> Integrity only
 MAC -> Integrity + Authentication
 Digital Signature -> Integrity + Authentication + Non-Repudiation

14.4 Cryptography Basics

- **Plaintext** → (**Encryption+Key**) → **Ciphertext** → (**Decryption+Key**) → **Plaintext**. **Cipher** = the algorithm (AES, DES, RSA...). **Key** = the secret value.
- **Kerckhoffs's Principle**: system must stay secure even if the algorithm is public — security depends on the **key**, not algorithm secrecy.
- **Cryptanalysis** = breaking a cipher without the key. **Cryptology** = **Cryptography** + **Cryptanalysis**.
- Cryptography gives Confidentiality/Integrity/Authentication/Non-Repudiation — **not Availability**.

14.5 Symmetric vs Asymmetric Cryptography

Feature	Symmetric	Asymmetric
Keys	1 shared secret key	2 (Public + Private)
Speed	Fast	Slow
Key distribution	Hard (must securely share)	Easy (public key can be open)
Main use	Bulk data encryption	Key exchange, Digital Signatures
Examples	DES, 3DES, AES, Blowfish	RSA, ElGamal, ECC, Diffie-Hellman

Number of unique symmetric keys needed for n users (full pairwise) = $n(n-1)/2$

- Symmetric disadvantages: key-distribution problem, poor scalability (formula above), **no non-repudiation** (either party could've created the message since both share the key).
- Asymmetric: a message encrypted with one key (of a pair) can only be decrypted with the *other* key of that same pair. Solves key distribution; enables digital signatures; but slower & unsuitable for large data.
- **Hash function**: fixed-length one-way output, NO key needed for basic integrity check (contrast with MAC which needs a key).

14.6 Encryption Algorithms

- **Block Cipher** (fixed-size blocks, e.g., DES/AES; needs padding) **vs Stream Cipher** (bit/byte-by-byte via XOR with a keystream, no padding, faster for real-time; keystream must NEVER repeat with same key). Examples stream: RC4, Salsa20, ChaCha20.

Feature	DES	3DES	AES
Block size	64 bits	64 bits	128 bits
Key size	56 bits (effective)	112/168 bits	128/192/256 bits
Security	Low (obsolete)	Medium (deprecated)	High (current standard)
Structure	16-round Feistel network	3x DES (Encrypt-Decrypt-Encrypt)	Substitution-Permutation

- **DES**: IBM, 1977 standard; 64-bit blocks, 56-bit key (64-bit key minus 8 parity bits); 16 Feistel rounds; broken by brute force → replaced by AES.
- **AES**: NIST selected 2001; 128-bit blocks; 128/192/256-bit keys; faster AND more secure than DES/3DES; current standard.

- **Modes of operation** (how a block cipher handles data > 1 block): **ECB, CBC, CFB, OFB, CTR**.
- **RSA**: asymmetric; security rests on difficulty of **factoring large primes**; used for encryption, digital signatures, key exchange, SSL/TLS.
- **Diffie-Hellman (DHKE)**: asymmetric — NOT an encryption algorithm, only a **key-exchange** method; shared secret is never transmitted; each side computes the same key independently; vulnerable to **Man-in-the-Middle** unless combined with authentication.

Feature	RSA	Diffie-Hellman
Purpose	Encryption & Digital Signature	Key Exchange only
Encrypts data?	Yes	No
Produces shared secret?	No	Yes
Provides authentication (by itself)?	Yes	No

Questions

Q1. Which security goal is NOT guaranteed by cryptography alone?

A. Confidentiality B. Integrity C. Availability D. Authentication

Answer: C. Why: Availability requires redundancy/anti-DoS infrastructure — encryption cannot stop resource exhaustion attacks.

Q2. An attacker resends a previously captured valid login message to gain access. This is a:

A. Masquerade Attack B. Replay Attack C. Traffic Analysis D. Modification Attack

Answer: B. Why: Reuse of a genuine, previously captured message defines Replay — the credentials/message are real, unlike Masquerade (stolen identity) or Modification (altered content).

Q3. [Numerical] In a symmetric-key system, 10 users need pairwise unique keys. How many keys are required?

A. 10 B. 45 C. 90 D. 100

Answer: B. Why: $n(n-1)/2 = 10 \times 9/2 = 45$.

Q4. Which mechanism provides Integrity + Authentication + Non-Repudiation, unlike a plain hash (integrity only) or MAC (integrity+authentication only)?

A. Symmetric encryption B. Digital Signature C. Simple checksum D. VPN

Answer: B. Why: Digital Signature uses asymmetric crypto (private key signs), so only the owner could have produced it — giving non-repudiation, which a shared-key MAC cannot provide.

Q5. [Trap] Traffic Analysis is classified as passive even though it yields useful intelligence, because:

A. It always uses unencrypted data B. Passive attacks are defined by absence of data modification, not by the value of information gained C. It targets availability D. It modifies packet headers

Answer: B. Why: Passive = no data altered; traffic analysis only observes metadata patterns (timing, size, frequency) without touching content.

Q6. Kerckhoffs's Principle states that a cryptosystem should remain secure even if:

A. The key is publicly known B. The algorithm is publicly known, provided the key stays secret C. Both algorithm and key are public D. No key is used at all

Answer: B. Why: Security must rest on the secrecy of the key alone, not on hiding the algorithm — a core assumption behind modern public cipher design (AES, RSA, etc.).

Q7. DES uses a 64-bit key, but its effective security strength comes from only:

A. 64 bits B. 56 bits C. 48 bits D. 32 bits

Answer: B. Why: 8 of the 64 bits are parity bits, leaving 56 effective key bits — this small size is why DES is now considered insecure (brute-forceable).

Q8. [Trap] Diffie-Hellman Key Exchange is best described as:

A. A symmetric block cipher B. An algorithm to encrypt bulk data C. A method to establish a shared secret key, not an encryption algorithm itself D. A digital signature scheme

Answer: C. Why: DHKE only lets two parties agree on a shared secret over an insecure channel; it does not encrypt data and provides no authentication by itself (vulnerable to MITM without added authentication).

Q9. Which cipher mode/type is preferred for real-time streaming data because it avoids padding and encrypts continuously?

A. Block cipher (ECB mode) B. Stream cipher C. AES with CBC D. RSA

Answer: B. Why: Stream ciphers XOR data with a keystream bit/byte-by-byte, needing no padding and suiting continuous, low-latency transmission — unlike block ciphers, which need full blocks (and possibly padding).

15. Hash/MAC/Digital Signature, Authentication & Security Tools

15.1 Hash Functions

- Converts any-length input → **fixed-length** digest; **one-way** (irreversible); Hash = H(M).
- Good hash properties:** fixed-length output, deterministic, one-way, fast computation, **avalanche effect** (tiny input change → completely different hash), **collision resistant**.
- Used for integrity checking (NOT confidentiality — hashing isn't encryption, there's no decryption step).

Algorithm	Output size	Status
MD5	128 bits	Insecure — vulnerable to collision attacks; legacy only
SHA-1	160 bits	Deprecated (collision attacks found)
SHA-256	256 bits	Recommended, widely used
SHA-384 / SHA-512	384 / 512 bits	Stronger variants
SHA-3	Variable	Modern, recommended

- TRAP:** MD5's 128-bit output and known collision vulnerabilities make it unsuitable for security-critical integrity checks today — a very common "which is insecure" question.

15.2 MAC vs HMAC vs Hash

Hash: Hash = H(M) → Integrity only, no key
 MAC: MAC = F(K, M) → Integrity + Authentication (shared secret key K)
 HMAC: Hash function + secret key → stronger, tamper-resistant MAC (commonly built on SHA-256/SHA-3)

Feature	Hash	MAC	HMAC
Uses a key?	No	Yes (shared secret)	Yes (shared secret + hash function)
Provides	Integrity only	Integrity + Authentication	Integrity + Authentication (more tamper-resistant)
Non-repudiation?	No	No (shared key ⇒ either side could forge)	No

15.3 Digital Signature

- Uses **asymmetric** crypto: sign with sender's **Private Key**, verify with sender's **Public Key** (opposite direction of normal encryption-for-confidentiality use of key pairs).
- Working:** Message → Hash → **Encrypt digest with sender's private key** = signature → send (message + signature) → receiver decrypts signature with sender's public key → receiver independently hashes the received message → **compare the two digests**; match ⇒ valid.
- Achieves: **Authentication, Integrity, Non-Repudiation**. Does **NOT by itself provide Confidentiality** (message itself isn't encrypted, only the digest is signed) — TRAP.

15.4 Digital Certificate & PKI

- Digital Certificate:** binds an identity to a Public Key; issued by a **Certificate Authority (CA)**; contains Subject, Public Key, Issuer, Serial Number, Validity Period, CA's own Digital Signature.
- CA:** trusted 3rd party that issues/verifies/manages certificates.
- PKI (Public Key Infrastructure):** the whole framework — CA, Registration Authority (RA), Certificates, Public/Private keys — that makes public-key crypto usable at scale (e.g., HTTPS).

15.5 Authentication Methods & Kerberos

- Password** → **OTP** (one-time, single-use) → **2FA** (exactly 2 factors, e.g., password+OTP) → **MFA** (2 or more factors).
- Kerberos:** network authentication protocol using **secret-key (symmetric) cryptography** + a trusted **KDC (Key Distribution Center)**; provides **mutual authentication**, avoids sending passwords over the network, enables **Single Sign-On (SSO)**. Used in Windows Active Directory.
- TRAP:** Kerberos relies on **symmetric** (secret-key) cryptography via a trusted third party — NOT on public-key/asymmetric crypto or digital certificates like PKI/CA.

15.6 Security Tools

Tool	Function	Blocks traffic?
Firewall	Filters traffic by rules (IP/port/protocol/application) at the network perimeter	Yes (by rule)
IDS	Detects and alerts on suspicious activity	No — passive, monitor-only
IPS	Detects AND automatically blocks malicious traffic in real time	Yes
VPN	Encrypted tunnel over a public network (Remote-Access or Site-to-Site)	N/A (confidentiality, not filtering)
SSL/TLS	Encrypted communication over the Internet (TLS = modern, more secure successor to SSL)	N/A

Firewall types: Packet Filtering (IP/port/protocol) → Stateful Inspection (tracks connection state) → Circuit-Level Gateway (verifies TCP handshake) → Application-Level Gateway/Proxy (inspects HTTP/FTP/SMTP payload) → Next-Gen Firewall (adds Deep Packet Inspection + Intrusion Prevention).

- **TRAP:** **IDS only alerts; it does NOT block traffic.** IPS is the active, blocking counterpart — this IDS-vs-IPS distinction is one of the most frequently tested facts in this topic.

Questions

Q1. Which hash algorithm is now considered insecure due to demonstrated collision attacks?

A. SHA-256 B. SHA-3 C. MD5 D. SHA-512

Answer: C. **Why:** MD5 (128-bit digest) has known practical collision attacks and is no longer recommended for security-critical use.

Q2. [Trap] What is the key difference between a MAC and a plain Hash function?

A. MAC produces a variable-length output B. MAC requires a shared secret key; plain hash needs no key C. Hash is reversible, MAC is not D. There is no difference

Answer: B. **Why:** MAC = $F(K, M)$ uses a secret key for authentication+integrity; a plain hash $H(M)$ needs no key and only verifies integrity.

Q3. A Digital Signature is created by encrypting the message digest with the sender's:

A. Public Key B. Private Key C. Shared secret key D. Session key

Answer: B. **Why:** Signing uses the sender's Private Key; anyone can then verify it using the corresponding Public Key — the reverse role of keys compared to confidentiality-focused asymmetric encryption.

Q4. [Trap] Does a Digital Signature by itself provide Confidentiality?

A. Yes, always B. No — only the digest is encrypted/signed, not the whole message C. Yes, but only for text messages D. No signature scheme provides Integrity either

Answer: B. **Why:** Digital signatures target Authentication, Integrity, and Non-Repudiation; the message body itself is sent as-is unless separately encrypted.

Q5. Which entity issues and digitally signs Digital Certificates to bind identities to public keys?

A. Registration Authority only B. Certificate Authority (CA) C. KDC D. IDS

Answer: B. **Why:** The CA is the trusted third party that issues, signs, and manages digital certificates within a PKI.

Q6. Kerberos achieves secure authentication primarily through:

A. Public-key certificates issued by a CA B. A trusted KDC using symmetric (secret-key) cryptography C. Digital signatures D. Firewalls

Answer: B. **Why:** Kerberos is built on secret-key cryptography and a trusted Key Distribution Center — distinct from PKI/CA-based asymmetric certificate systems.

Q7. [Trap] Which security tool only detects and alerts on suspicious activity but does NOT block it?

A. Firewall B. IPS C. IDS D. VPN

Answer: C. **Why:** IDS = Intrusion *Detection* System — passive monitoring/alerting only; IPS additionally blocks malicious traffic automatically.

Q8. A firewall that inspects application-layer payloads such as HTTP or FTP commands is called a:

A. Packet Filtering Firewall B. Circuit-Level Gateway C. Application-Level Gateway (Proxy Firewall) D. Stateful Inspection Firewall

Answer: C. **Why:** Only Application-Level Gateways (proxy firewalls) inspect payload content at the application layer; the others work at IP/port or connection-state level.

Q9. SSL and TLS are best described as:

A. Firewall types B. Encryption algorithms for disk storage only C. Protocols providing encrypted communication over the Internet, with TLS as the modern successor to SSL D. Authentication factors like OTP

Answer: C. **Why:** SSL/TLS secure data in transit (e.g., HTTPS); TLS supersedes the older, weaker SSL while serving the same purpose.